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Interventions for improving treatment outcomes in adolescents on antiretroviral therapy with unsuppressed viral loads: a systematic review protocol

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Interventions for improving treatment outcomes in Adolescents on antiretroviral therapy with unsuppressed viral loads: a systematic review protocol

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Abstract

Introduction: Adolescents represent one of the most underserved population groups among people living with HIV. With successes in the Prevention of Mother to Child transmission programs and advances in pediatric HIV treatment programs, a large population of HIV-infected children are surviving into adolescence. Adolescence presents unique challenges that increases the risk of non-suppressed viral loads in adolescents living with HIV (ALHIV). There is need to develop, implement and test interventions to improve viral load suppression among ALHIV. Systematic reviews of recent studies present scarce and inconclusive evidence of effectiveness of current interventions, especially for unsuppressed adolescents. This systematic review protocol provides a description of a planned review of interventions to improve treatment outcomes among unsuppressed ALHIV.

Methods and analysis: A comprehensive search string will be used to search six bibliographic databases: PubMed/MEDLINE, Sabinet, EBSCOhost, CINAHL, Scopus and ScienceDirect, for relevant studies published between 2010-2020. Identified articles will be exported into Mendeley Reference Management software and two reviewers will independently screen the titles, abstracts and full texts for eligibility. Any discrepancies between the two reviewers will be resolved by a third reviewer. Studies reporting on interventions to improve retention, adherence and viral load suppression for adolescents with unsuppressed viral loads will be considered for inclusion. The systematic review will be performed and reported according to the Preferred Reporting Items for Systematic reviews and Meta-Analyses Protocols. Where feasible, the meta-analysis will be conducted using Stata Statistical Software: Release 16. College Station, TX: StataCorp LLC.

Ethics and dissemination: The systematic review entails abstracting and reviewing already publicly available data rather than any involvement of participants, therefore no ethical clearance will be required. Results will be shared with relevant policy makers, program managers and service providers. We also plan to publish the results, and share the findings using platforms such as conferences and webinars.

PROSPERO registration number: Submitted for registration (ID 232440)

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Strengths and limitations of this study

- This study contributes to the knowledge around available and evaluated interventions targeting adolescents with unsuppressed viral loads that could inform adolescent-friendly HIV services guidelines and policy in both health facilities and community settings.
- To our knowledge, this will be the first systematic review to focus on targeted interventions to improve viral load suppression specifically for adolescents with unsuppressed viral loads.
- This protocol utilizes the standardized Preferred Reporting Items for Systematic reviews and Meta-Analyses Protocols 2015 guidelines.
- The search strategy utilizes six major electronic databases which will include peer-reviewed and grey literature.
- A potential limitation of this study includes the limitation to studies published in English language only, which has the potential to miss other relevant studies published in other languages.

Introduction

Due to the successes in prevention of mother-to-child transmission (PMTCT) programs worldwide and advances in pediatric HIV treatment, children with HIV are surviving to reach adolescence (1). Innovations for Early Infant Diagnosis such as Point of Care diagnostics and prompt initiation on life-saving HIV treatment, has meant a large population of perinatally HIV-infected children are growing into adolescence (2). Worldwide, an estimated 1.6 million adolescents between 10-19 years were living with HIV in 2018; with nearly 85% living in sub-Saharan Africa (3, 4). The World Health Organization (WHO) defines adolescents as children or young adults between 10-19 years of age (5). Although data on HIV treatment among adolescents is scarce, access and uptake of comprehensive care and treatment packages among this vulnerable population group is reportedly lower than adults (6). In 2019, only 53% (36-64%) of children aged 0-14 years were accessing treatment as compared to 68% (54-80%) coverage among 15 years and older individuals living with HIV (7). In most sub-Saharan Africa countries, public health facilities are ill-equipped to give guidance and support for adolescents living with HIV (ALHIV) to remain engaged in HIV care and adhere to medication regimens (8).

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Good adherence to treatment regimens and consistent engagement in HIV care is vital for achieving sustained viral suppression, which is essential for ensuring the wellbeing of ALHIV and reducing chances of onward HIV transmission (9). Poor ART adherence increases the risk of viral drug-resistance, limits treatment efficacy, leading to disease progression, and reduces future therapeutic options as well as increasing the risk of transmission due to unsuppressed viral replication (10). Although reported ART adherence is high globally (>95%), concerns have been raised about waning adherence over time including loss of patients from HIV programs when scaling up (11). Evidence-based interventions to address adherence challenges for people on ART include individual and group adherence counselling, mHealth platforms, community and home-based strategies, pharmacist counselling and monitoring, task-shifting, medication fast-tracking, nutrition support and provision of disability grants (12). For ALHIV, individual counselling, group counselling and peer support, such as in teen clubs, have been some of the most common interventions.

Adolescents living with HIV have unique needs and are notably underserved globally and in national responses, which negatively affects their access to ART and results in poor ART adherence and inferior treatment outcomes such as achieving and maintain virologic suppression (13). According to WHO, a maintained viral load of <1000 ribonucleic acid (RNA) copies per ml of plasma is considered evident of virologic suppression (14). However recent debates have suggested reducing the suppression threshold to 200 copies/ml, informed by evidence from HPTN 052 findings, PARTNERS and Opposites Attract, who all reported that successful ART with viral suppression prevented transmission when viral load was undetectable below 200 copies/mL (15).

A 2014 systematic review showed global adherence estimates ranging from 16% to 99% among adolescents whilst a meta-analysis of data for young adults and adolescents aged 12–24 years in 53 countries found an estimated pooled adherence at 84% in Asia and Africa. Rates of viral suppression (< 400 copies/mL) ranged from 52% to 87% in Asia, 27% to 89% in Africa, and from 37.5% to 49% in Central and South America (16). A recent review of eleven studies done in sub-Saharan Africa identified a complex web of 44 barriers and 29 facilitators impacting ART adherence among adolescents, with stigma, lack of assistance, ART side-effects and forgetfulness as top barriers whilst top facilitators were caregiver support, knowledge of status and peer support

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groups (17). The barriers and facilitators of ART adherence among adolescents in low- and middle-income countries have been noted as family structure, a burdensome ART regimen, attitudes about medication and various healthcare system factors (18).

Despite recent increased focus and several systematic reviews synthesizing interventions to improve treatment outcomes among adolescents, we could not find reviews specifically focusing on studies targeting unsuppressed adolescents. Previous reviews have identified specific types of interventions that show promise for improved adherence to ART and retention in HIV care with very limited quantity of studies and low quality in several studies focusing on adolescents aged 10-19 years published up until 2015 (19). A review by MacPherson *et al*. focused on studies published between 2001 and 2014 evaluating effectiveness of interventions on service delivery to improve linkage to treatment initiation, retention in HIV care, and ART adherence among ALHIV aged 10-19 years (13). Considering that only 11 studies were included in the study, with low-to-moderate methodological quality, from high-income countries and none from countries with a generalized HIV epidemic, the authors suggested further investigations on their findings which included promising results from individual and group education and counseling, increased accessibility to clinics, youth-friendly services and financial incentives (13).

Casale *et al.* reviewed studies published between January 2016 and June 2018, and provided an update on evidence from empirical evaluations on interventions designed to enhance ART adherence and retention in care for adolescents aged 10-19 years and youth aged 15-24 years living with HIV (19). The authors reported on 10 relevant studies, mostly from sub-Saharan Africa, utilizing six clinic-level interventions, three community or household level intervention, and one mHealth trial at individual level, and they concluded that there is need to further develop and test multifaceted interventions beyond facility-based interventions, that address broader socio-economic barriers, prioritizing adolescents aged 10-19 years, if we are to address the treatment gap among this population group (19). A more recent review of articles published between September 2015 and January 2019 by Lindsey *et al.* reported that ART adherence interventions for adolescents and youth implemented in low- and middle-income countries showed mixed results from interventions at patient-level and health services level, with none of the interventions improving ART adherence or viral load suppression (20). The authors argued that intensive home-

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based case management models of care hold much promise in improving treatment outcomes in adolescents and youth and warrant further research (20).

Rationale for the review

Globally, it is apparent that there has been an increased focus in addressing the treatment gap among ALHIV, and in recent years, there have been a number of systematic reviews focusing on interventions to improve ART adherence and retention in HIV care among adolescents and young adults in general. While the gaps described above and the systematic reviews cited present opportunities and specific interventions that show promise, they also highlight the limited quantity and low-to-moderate quality of studies and evidence on interventions targeting ALHIV. These reviews commonly include interventions for all adolescents and youth, despite their viral load suppression levels, which include clinic-level interventions, individual-level, and community- or household-level interventions. Considering the urgent need to design, implement, and test interventions to keep adolescents in care and achieve and maintain viral load suppression, we propose to review studies that specifically evaluated interventions targeting unsuppressed adolescents on HIV treatment. This review will focus on unsuppressed adolescents to provide an updated data synthesis of all adherence interventions to close the treatment gap, specifically targeting unsuppressed adolescents. We aim to broaden the synthesized knowledge base to include new and innovative interventions for this particular population, and potentially reinforce or modify already existing promising interventions for achievement of viral re-suppression among unsuppressed adolescents.

Methods

This systematic review will use the seven steps as described by Eggar, Davey and Smith (22). The seven steps are outlined as (1) formulate the review question; (2) define the inclusion and exclusion criteria; (3) develop a search strategy; (4) study selection; (5) assess the quality of studies; (6) extract data, and (7) analyse or synthesise the data.

Review question

The following questions will guide the review of the literature published between 2010 and 2020:

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1. What interventions are being implemented to improve treatment outcomes for adolescents on ART with unsuppressed viral loads?
2. How effective have the interventions been in achieving better viral load suppression levels among unsuppressed adolescents?

Inclusion and exclusion criteria

The literature search will be guided by the following inclusion criteria:

1. Published studies in peer-reviewed journals and grey literature
2. The study *population* includes ALHIV aged 10-19 years
3. Reported the association between the *intervention*, and retention, adherence and viral load suppression
4. Reported *comparisons* in quantitative measures of retention, adherence and viral load suppression
5. Study *outcomes* include retention, adherence and viral load suppression
6. Studies published in English language
7. Studies from all countries published between 2010 and 2020.

Studies that are not available in English language will be excluded, as well as review papers, conference abstracts, presentations, editorials and letters.

The decision to only include studies reported in English language was necessitated by the limitations in capacity and resources among the review team, which restricts the review to English language only. Studies may include younger children below 10 years or young adults over 19 years, and justification for inclusion will be provided if these are included in the analysis. Additionally, other studies may focus on adherence interventions and viral load suppression for all people living with HIV, including adolescents in the study population or sample.

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Table 1. PICOT based inclusion criteria

Patient population	Adolescents aged 10-19 years living with HIV, acquired perinatally or behaviourally, who have unsuppressed viral loads.
Interventions	Any group-based or individual-level interventions to improve treatment outcomes targeting adolescents who have unsuppressed viral loads, including Differentiated Service Delivery (DSD) models.
Comparisons	Studies may report comparisons on quantitative measures of retention, adherence and viral load suppression between intervention and standard care.
Outcomes	Primary outcome - viral load suppression Secondary outcomes – retention in care and ART adherence
Time	2010 – 2020
Other considerations	Language: English
	Study designs: Quantitative and Mixed methods
	Geographical area: Global

Outcome measures

ART adherence – adherence can be measured using pill counts, pharmacy refill data, self-reporting, as well as physician/nurse assessments and appointment keeping. More robust measures such as Medical Events Monitoring System (MEMS) and other new technologies will all be considered in this review. Optimal adherence will be considered to be at ≥ 95% of prescribed medication doses taken per specified period.

Retention in HIV care – methods of measuring retention in care may be based on missed clinic visits or based on kept clinic visits. Retention variables of interest may also include whether patient is alive and on continued ART, defaulted/lost, lost to follow-up (LTFU), transferred-out or died.

Viral load suppression – although WHO viral load suppression threshold in 1000 copies/mL, more recent studies may have used lower levels of viral load counts to define viral load suppression or

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disaggregate low level viremia and suppression. All definitions of viral load suppression will be considered in this review.

Search strategy for identification of studies

The search strategy was developed with the assistance of the faculty Librarian who will provide further guidance and advice in conducting the searches for the review. A comprehensive database search of literature reporting on treatment interventions and viral load suppression will be systematically conducted in six major databases: PubMed/MEDLINE, Sabinet, EBSCOhost, CINAHL, Scopus and ScienceDirect. We will search the identified databases for all articles published between 2010 and 2020 that meet the specified selection criteria. The search syntax will first be used to perform a database search on PubMed/MEDLINE to test and optimize the search strategy, which will subsequently be replicated and adapted as needed for other database searches. Full-text articles will be sourced using a list of key words and terms for the search strategy using the "AND" and "OR" Boolean operators as appropriate. The search terms will include "Adolescents", "antiretroviral therapy", "retention", "adherence", "viral load", "suppression", "unsuppressed". Other terms such as "non-suppression", "HIV", "virologic", "adherence intervention" will also be included.

Study selection

Studies will be included in the systematic review using the Population/Patient/Problem, Intervention, Comparison, Outcome and Time (PICOT) mnemonics. We will also consider studies done using quantitative or mixed methods, with a worldwide search without any particular geographical location limitations. Restricting reviews to a particular study design, such as RCTs only, may restrict findings to effectiveness only rather than seeking answers to more complex phenomena such as why an intervention may or may not be effective (23).

The selection, screening and reporting of this review will be done in accordance with the reporting guidance provided in the Preferred Reporting Items for Systematic Reviews and Meta-Analyses Protocol (PRISMA-P) 2015 guidelines (21). Selected studies will be collated and exported into Mendeley Reference Management Software, which will be used to manage the citations and recording the number of hits for each article from all databases, as well as removing duplicate

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records (24). Two reviewers will independently review titles and abstracts of all articles retrieved to assess eligibility for inclusion. The two reviewers will further review full-texts of the selected articles independently, to establish compliance with the eligibility criteria for inclusion in the final assessment (22). Any disagreements on inclusion or exclusion of any article will be referred to a third reviewer authorized to resolve any disagreements for inclusion of articles. A PRISMA flow diagram will be presented to describe the study selection decisions made.

Data Extraction & Quality assessment

Conducting a quality assessment may determine the inclusion or exclusion of an initially selected study, and this should consider the robustness (validity and reliability) of findings of a particular study (23). We will create a data extraction matrix in Microsoft Excel to capture study characteristics and outcomes data, such as Author(s), study year and duration, publication year, geographical location, target population, sample, purpose, study design, intervention, data collection and analyses, outcomes, results, limitations and conclusions. Only the data reported in the articles will be extracted and used in the quality assessment although additional information may be sought from authors for clarification.

Data analysis/synthesis: Assessment of evidence using the GRADE system

The two primary review authors will independently assess the selected articles using the GRADE (grading of recommendations assessment, development, and evaluation) considerations to downgrade or upgrade the certainty of evidence on the interventions and the specific outcomes reported by each study. Independent conclusions will be drawn on how the interventions compare to each other, on the specified treatment outcomes considering the estimates of effects, and the certainty of the evidence (risk of bias, confidence in evidence/ consistency in effects, quality of evidence, imprecision, indirectness, and publication bias) (25, 26). Since the systematic review will focus on unsuppressed ALHIV, we will consider outcomes reported on adolescents who may have had adherence challenges. Studies which included clinical or medication issues such as drug resistance may be considered with appropriate justifications.

We will conduct a meta-analysis only if the interventions, research questions or focus, and study participants allow for a meaningful meta-analysis. The meta-analysis will be conducted using Stata

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Statistical Software: Release 16. College Station, TX: StataCorp LLC. The X^2 test and the Higgins I^2 statistic will be used to quantify statistical heterogeneity by estimating the variability between study estimates, with I^2 statistic values of 25%, 50%, and 75% representing low, medium and high heterogeneity respectively (27). Forest plots will be used to visualize the extent of heterogeneity or homogeneity among the selected studies.

Publication bias described as the tendency by authors to report positive findings, will be presented and visualized on a funnel plot, as well as statistically assessed using the Harbord`s regression test to assess the asymmetry of the funnel plot (28, 29). The contour-enhanced funnel plot will be used to distinguish between asymmetry caused by publication bias versus other causes, which may not have been successfully accomplished by the standard method (30). If it is not possible to do a meta-analysis, the findings extracted onto the Excel spreadsheet described above will be summarized and reported in a table form presenting key components of the studies. Subgroup analysis may include facility vs community-based interventions and younger vs older adolescents.

Patient and Public involvement

No patients or public population is involved in the development and conducting of this study, as this is a protocol for a systematic review of already available literature.

Ethics and dissemination

The systematic review entails abstracting and reviewing already publicly available data rather than any involvement of participants, therefore no ethical clearance or approval will be required.

We plan to share results of this review with relevant policy makers and guidelines developers, program managers and service providers, particularly providing evidence-based guidance for interventions targeting unsuppressed adolescents. We also plan to publish the results of the review, as well as sharing the findings with key stakeholders using platforms such as conferences, and presentations through webinars and the HIV Project ECHO (Extension for Community Health Outcomes) (31).

Contributions: FKM conceived the idea, developed the research questions and study methods and contributed meaningfully to the drafting and editing, he also approved the final manuscript. BvW

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aided in developing the research questions and study methods, contributed meaningfully to the drafting and editing, and approved the final manuscript.

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References

1. Bekker L-G, Siberry GK, Hirnschall G. Ensuring Children and Adolescents Are Not Left Behind [Internet]. 2018 [cited 2020 May 6]. Available from: https://journals.lww.com/jaids

2. UNICEF. HIV and AIDS in Adolescents - UNICEF Data [Internet]. 2020 [cited 2020 Dec 11]. Available from: https://data.unicef.org/topic/adolescents/hiv-aids/

3. UNICEF. HIV and AIDS in Adolescents - UNICEF Data:Turning the tide against AIDS will require more concentrated focus on adolescents and young people [Internet]. 2019 [cited 2020 May 6]. Available from: https://data.unicef.org/topic/adolescents/hiv-aids/

4. UNAIDS. UNAIDS data 2019 [Internet]. 2019 [cited 2020 May 6]. Available from: https://www.unaids.org/sites/default/files/media_asset/2019-UNAIDS-data_en.pdf

5. WHO. WHO | Definition of key terms. WHO [Internet]. 2013 [cited 2016 Oct 11]; Available from: http://www.who.int/hiv/pub/guidelines/arv2013/intro/keyterms/en/

6. UNAIDS. Ending the AIDS epidemic for adolescents, with adolescents: A practical guide to meaningfully engage adolescents in the AIDS response. Geneva; 2016.

7. UNAIDS. Global HIV & AIDS statistics — 2020 fact sheet | UNAIDS [Internet]. 2020 [cited 2020 Dec 12]. Available from: https://www.unaids.org/en/resources/fact-sheet

8. Ssali L, Kalibala S, Birungi J, Egessa A, Wangisi J, Lyavala J, et al. Retention of adolescents living with HIV in care, treatment, and support programs in Uganda. 2014 [cited 2018 Feb 27]; Available from: http://www.hivcore.org/Pubs/Uganda_AdolHAART_Rprt.pdf

9. Grimsrud AT, Pike C, Bekker LG. The power of peers and community in the continuum of HIV care [Internet]. Vol. 8, The Lancet Global Health. Elsevier Ltd; 2020 [cited 2020 Nov 16]. p. e167–8. Available from: https://pubmed.ncbi.nlm.nih.gov/31924538/

10. Kim S-H, Gerver SM, Fidler S, Ward H. Adherence to antiretroviral therapy in adolescents living with HIV: systematic review and meta-analysis. AIDS [Internet]. 2014

For peer review only

Aug 24 [cited 2018 Feb 27];28(13):1945–56. Available from: http://www.ncbi.nlm.nih.gov/pubmed/24845154

11. Nachega JB, Mills EJ, Schechter M. Antiretroviral therapy adherence and retention in care in middle-income and low-income countries: current status of knowledge and research priorities. Curr Opin HIV AIDS [Internet]. 2010 Jan [cited 2018 Sep 30];5(1):70–7. Available from: https://insights.ovid.com/crossref?an=01222929-201001000-00011

12. Ridgeway K, Dulli LS, Murray KR, Silverstein H, Santo LD, Olsen P, et al. Interventions to improve antiretroviral therapy adherence among adolescents in low- and middle-income countries: A systematic review of the literature [Internet]. Vol. 13, PLoS ONE. 2018. 1–33 p. Available from: http://dx.doi.org/10.1371/journal.pone.0189770

13. MacPherson P, Munthali C, Ferguson J, Armstrong A, Kranzer K, Ferrand RA, et al. Service delivery interventions to improve adolescents' linkage, retention and adherence to antiretroviral therapy and HIV care. Trop Med Int Health [Internet]. 2015 Aug [cited 2016 Sep 30];20(8):1015–32. Available from: http://www.ncbi.nlm.nih.gov/pubmed/25877007

14. WHO. WHO | Viral suppression after 12 months of antiretroviral therapy in low- and middle-income countries: a systematic review. WHO [Internet]. 2013 [cited 2017 Jul 13]; Available from: http://www.who.int/bulletin/volumes/91/5/12-112946/en/

15. WHO. WHO | Viral suppression for HIV treatment success and prevention of sexual transmission of HIV [Internet]. 2018 [cited 2020 Dec 12]. Available from: https://www.who.int/hiv/mediacentre/news/viral-supression-hiv-transmission/en/

16. Ridgeway K, Dulli LS, Murray KR, Silverstein H, Santo LD, Olsen P, et al. Interventions to improve antiretroviral therapy adherence among adolescents in low- and middle-income countries: A systematic review of the literature [Internet]. Vol. 13, PLoS ONE. Public Library of Science; 2018 [cited 2020 Jul 9]. Available from: /pmc/articles/PMC5749726/?report=abstract

17. Ammon N, Mason S, Corkery JM. Factors impacting antiretroviral therapy adherence among human immunodeficiency virus-positive adolescents in Sub-Saharan Africa: a systematic review. Public Health [Internet]. 2018 Apr 1 [cited 2018 Sep 30];157:20–31.

For peer review only

Available from: http://www.ncbi.nlm.nih.gov/pubmed/29501984

18. Hudelson C, Cluver L. Factors associated with adherence to antiretroviral therapy among adolescents living with HIV/AIDS in low- and middle-income countries: a systematic review. AIDS Care [Internet]. 2015 [cited 2016 Sep 30];27(7):805–16. Available from: http://www.ncbi.nlm.nih.gov/pubmed/25702789

19. Casale M, Carlqvist A, Cluver L. Recent Interventions to Improve Retention in HIV Care and Adherence to Antiretroviral Treatment Among Adolescents and Youth: A Systematic Review. AIDS Patient Care STDS [Internet]. 2019 Jun 1 [cited 2020 Apr 26];33(6):237–52. Available from: https://www.liebertpub.com/doi/10.1089/apc.2018.0320

20. Reif LK, Abrams EJ, Arpadi S, Elul B, McNairy ML, Fitzgerald DW, et al. Interventions to Improve Antiretroviral Therapy Adherence Among Adolescents and Youth in Low- and Middle-Income Countries: A Systematic Review 2015–2019 [Internet]. Vol. 1, AIDS and Behavior. Springer; 2020 [cited 2020 Jul 31]. p. 1. Available from: /pmc/articles/PMC7223708/?report=abstract

21. Moher D, Shamseer L, Clarke M, Ghersi D, Liberati A, Petticrew M, et al. Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015 statement. Rev Esp Nutr Humana y Diet [Internet]. 2016 [cited 2020 Nov 3];20(2):148–60. Available from: https://pubmed.ncbi.nlm.nih.gov/25554246/

22. Egger M, Smith GD. Principles of and Procedures for Systematic Reviews. In: Systematic Reviews in Health Care: Meta-Analysis in Context: Second Edition. Wiley Blackwell; 2008. p. 23–42.

23. Grant MJ, Booth A. A typology of reviews: An analysis of 14 review types and associated methodologies [Internet]. Vol. 26, Health Information and Libraries Journal. John Wiley & Sons, Ltd; 2009 [cited 2020 Nov 3]. p. 91–108. Available from: https://onlinelibrary.wiley.com/doi/full/10.1111/j.1471-1842.2009.00848.x

24. Price A, Price, Amy. Mendeley and More for Systematic Reviews. Int Netw Knowl about Wellbeing [Internet]. 2016 Apr 26 [cited 2020 Nov 3]; Available from: http://www.ithinkwell.org/mendeley-and-more-for-systematic-reviews/

For peer review only

25. Mearns H, Otiku PK, Shelton M, Kredo T, Kagina BM, Schmidt BM. Screening strategies for adults with type 2 diabetes mellitus: A systematic review protocol [Internet]. Vol. 9, Systematic Reviews. BioMed Central; 2020 [cited 2020 Nov 16]. p. 156. Available from: https://systematicreviewsjournal.biomedcentral.com/articles/10.1186/s13643-020-01417-3

26. Brignardello-Petersen R, Izcovich A, Rochwerg B, Florez ID, Hazlewood G, Alhazanni W, et al. GRADE approach to drawing conclusions from a network meta-analysis using a partially contextualised framework. BMJ [Internet]. 2020 Nov 10 [cited 2020 Nov 16];371:m3907. Available from: https://www.bmj.com/lookup/doi/10.1136/bmj.m3907

27. Mengesha MM, Ajema D, Teshome A, Tura AK. The association between diagnosis disclosure and adherence to antiretroviral therapy among adolescents living with HIV in sub-Saharan Africa: A protocol for systematic review and meta-analysis [Internet]. Vol. 9, Systematic Reviews. BioMed Central; 2020 [cited 2020 Nov 7]. p. 160. Available from: https://systematicreviewsjournal.biomedcentral.com/articles/10.1186/s13643-020-01420-8

28. Egger M, Smith GD, Schneider M, Minder C. Bias in meta-analysis detected by a simple, graphical test. Br Med J [Internet]. 1997 Sep 13 [cited 2020 Nov 16];315(7109):629–34. Available from: https://www.bmj.com/content/315/7109/629

29. Rao G, Lopez-Jimenez F, Boyd J, D'Amico F, Durant NH, Hlatky MA, et al. Methodological standards for meta-analyses and qualitative systematic reviews of cardiac prevention and treatment studies a scientific statement from the American Heart Association. Circulation [Internet]. 2017 Sep 1 [cited 2020 Nov 16];136(10):e172–94. Available from: https://pubmed.ncbi.nlm.nih.gov/28784624/

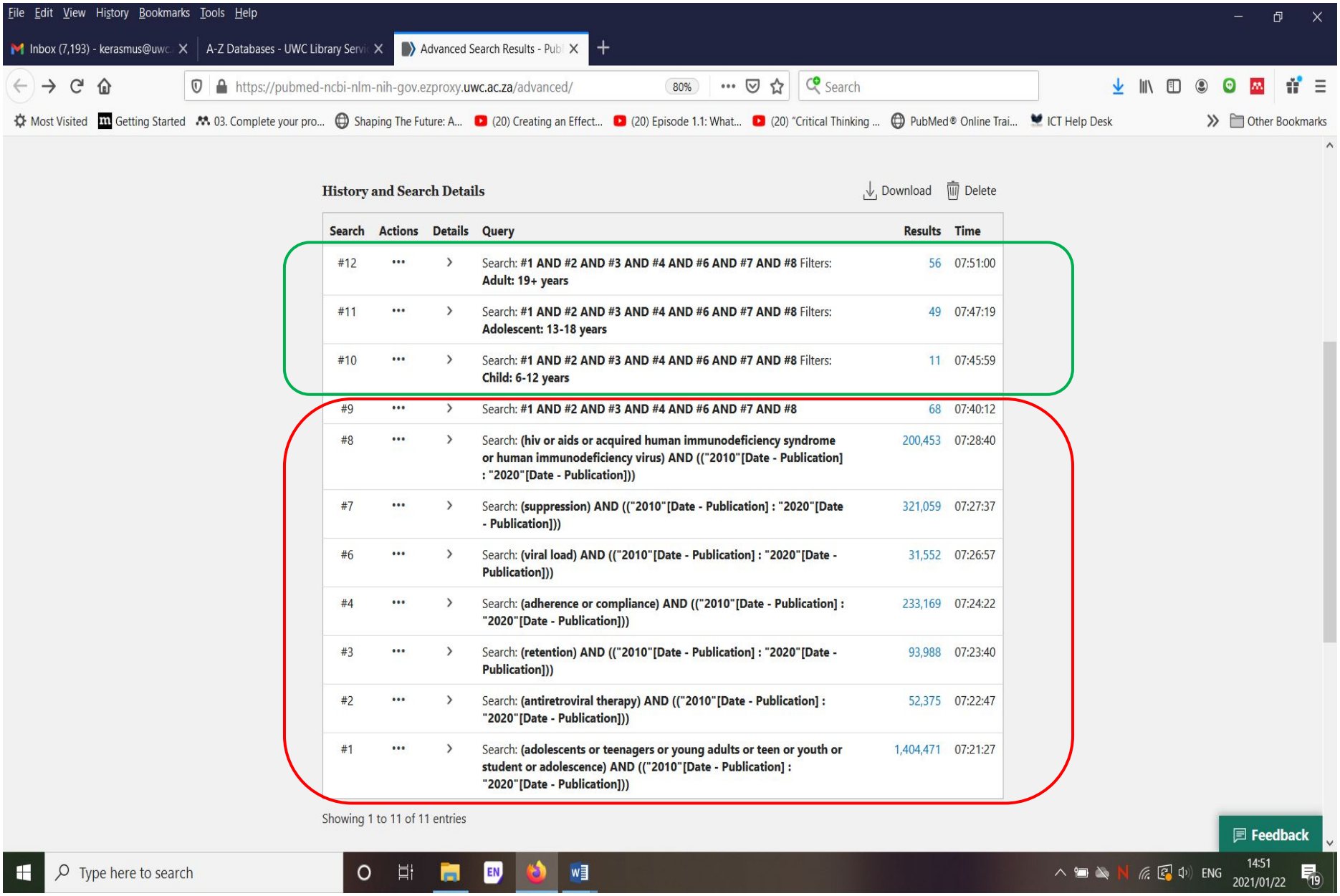
30. Chaimani A, Mavridis D, Salanti G. A hands-on practical tutorial on performing meta-analysis with Stata. Evid Based Ment Health [Internet]. 2014 Nov 1 [cited 2020 Nov 16];17(4):111–6. Available from: https://pubmed.ncbi.nlm.nih.gov/25288685/

31. Bikinesi L, O'Bryan G, Roscoe C, Mekonen T, Shoopala N, Mengistu AT, et al. Implementation and evaluation of a Project ECHO telementoring program for the Namibian HIV workforce. Hum Resour Health [Internet]. 2020 Sep 1 [cited 2020 Nov 17];18(1):1–10. Available from: https://link.springer.com/articles/10.1186/s12960-020-

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#12	•••	›	Search: **#1 AND #2 AND #3 AND #4 AND #6 AND #7 AND #8** Filters: **Adult: 19+ years**	56	07:51:00
#11	•••	›	Search: **#1 AND #2 AND #3 AND #4 AND #6 AND #7 AND #8** Filters: **Adolescent: 13-18 years**	49	07:47:19
#10	•••	›	Search: **#1 AND #2 AND #3 AND #4 AND #6 AND #7 AND #8** Filters: **Child: 6-12 years**	11	07:45:59
#9	•••	›	Search: **#1 AND #2 AND #3 AND #4 AND #6 AND #7 AND #8**	68	07:40:12
#8	•••	›	Search: **(hiv or aids or acquired human immunodeficiency syndrome or human immunodeficiency virus) AND (("2010"[Date - Publication] : "2020"[Date - Publication]))**	200,453	07:28:40
#7	•••	›	Search: **(suppression) AND (("2010"[Date - Publication] : "2020"[Date - Publication]))**	321,059	07:27:37
#6	•••	›	Search: **(viral load) AND (("2010"[Date - Publication] : "2020"[Date - Publication]))**	31,552	07:26:57
#4	•••	›	Search: **(adherence or compliance) AND (("2010"[Date - Publication] : "2020"[Date - Publication]))**	233,169	07:24:22
#3	•••	›	Search: **(retention) AND (("2010"[Date - Publication] : "2020"[Date - Publication]))**	93,988	07:23:40
#2	•••	›	Search: **(antiretroviral therapy) AND (("2010"[Date - Publication] : "2020"[Date - Publication]))**	52,375	07:22:47
#1	•••	›	Search: **(adolescents or teenagers or young adults or teen or youth or student or adolescence) AND (("2010"[Date - Publication] : "2020"[Date - Publication]))**	1,404,471	07:21:27

Showing 1 to 11 of 11 entries

Search	Actions	Details	Query	Results	Time
#12					
Search: **#1 AND #2 AND #3 AND #4 AND #6 AND #7 AND #8** Filters: **Adult: 19+ years**				56	07:51:00
#11					
~~Search: #1 AND #2 AND #3 AND #4 AND #6 AND #7 AND #8 Filters: Adolescent: 13-18 years~~				~~49~~	~~07:47:19~~
#10					
Search: **#1 AND #2 AND #3 AND #4 AND #6 AND #7 AND #8** Filters: **Child: 6-12 years**				11	07:45:59
#9					
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Reporting checklist for protocol of a systematic review.

Based on the PRISMA-P guidelines.

Instructions to authors

Complete this checklist by entering the page numbers from your manuscript where readers will find each of the items listed below.

Your article may not currently address all the items on the checklist. Please modify your text to include the missing information. If you are certain that an item does not apply, please write "n/a" and provide a short explanation.

Upload your completed checklist as an extra file when you submit to a journal.

In your methods section, say that you used the PRISMA-Preporting guidelines, and cite them as:

Moher D, Shamseer L, Clarke M, Ghersi D, Liberati A, Petticrew M, Shekelle P, Stewart LA. Preferred Reporting Items for Systematic Review and Meta-Analysis Protocols (PRISMA-P) 2015 statement. Syst Rev. 2015;4(1):1.

		Reporting Item	Page Number
Title			
Identification	#1a	Identify the report as a protocol of a systematic review	1
Update	#1b	If the protocol is for an update of a previous systematic review, identify as such	N/A
Registration			
	#2	If registered, provide the name of the registry (such as PROSPERO) and registration number	2
Authors			
Contact	#3a	Provide name, institutional affiliation, e-mail address of all protocol authors; provide physical mailing address of corresponding author	1
Contribution	#3b	Describe contributions of protocol authors and identify the guarantor of the review	11
Amendments			

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	#4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments	N/A
Support			
Sources	#5a	Indicate sources of financial or other support for the review	N/A
Sponsor	#5b	Provide name for the review funder and / or sponsor	N/A
Role of sponsor or funder	#5c	Describe roles of funder(s), sponsor(s), and / or institution(s), if any, in developing the protocol	N/A
Introduction			
Rationale	#6	Describe the rationale for the review in the context of what is already known	6
Objectives	#7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	7
Methods			
Eligibility criteria	#8	Specify the study characteristics (such as PICO, study design, setting, time frame) and report characteristics (such as years considered, language, publication status) to be used as criteria for eligibility for the review	7-8
Information sources	#9	Describe all intended information sources (such as electronic databases, contact with study authors, trial registers or other grey literature sources) with planned dates of coverage	9
Search strategy	#10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated	N/A
Study records - data management	#11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	9-11
Study records - selection process	#11b	State the process that will be used for selecting studies (such as two independent reviewers) through each phase of the review (that is, screening, eligibility and inclusion in meta-analysis)	9-10
Study records - data collection process	#11c	Describe planned method of extracting data from reports (such as piloting forms, done independently, in duplicate), any processes for	10

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		obtaining and confirming data from investigators	
Data items	#12	List and define all variables for which data will be sought (such as PICO items, funding sources), any pre-planned data assumptions and simplifications	8
Outcomes and prioritization	#13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	8
Risk of bias in individual studies	#14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	10
Data synthesis	#15a	Describe criteria under which study data will be quantitatively synthesised	10-11
Data synthesis	#15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data and methods of combining data from studies, including any planned exploration of consistency (such as I2, Kendall's τ)	10-11
Data synthesis	#15c	Describe any proposed additional analyses (such as sensitivity or subgroup analyses, meta-regression)	11
Data synthesis	#15d	If quantitative synthesis is not appropriate, describe the type of summary planned	11
Meta-bias(es)	#16	Specify any planned assessment of meta-bias(es) (such as publication bias across studies, selective reporting within studies)	11
Confidence in cumulative evidence	#17	Describe how the strength of the body of evidence will be assessed (such as GRADE)	10

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BMJ Open

Interventions for improving treatment outcomes in adolescents on antiretroviral therapy with unsuppressed viral loads: a systematic review protocol

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Interventions for improving treatment outcomes in Adolescents on antiretroviral therapy with unsuppressed viral loads: a systematic review protocol

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Abstract

Introduction: Adolescents represent one of the most underserved population groups among people living with HIV. With successes in the Prevention of Mother to Child transmission programs and advances in pediatric HIV treatment programs, a large population of HIV-infected children are surviving into adolescence. Adolescence presents unique challenges that increases the risk of non-suppressed viral loads in adolescents living with HIV (ALHIV). There is need to develop, implement and test interventions to improve viral suppression among ALHIV. Systematic reviews of recent studies present scarce and inconclusive evidence of effectiveness of current interventions, especially for adolescents. This protocol provides a description of a planned review of interventions to improve treatment outcomes among ALHIV.

Methods and analysis: A comprehensive search string will be used to search six bibliographic databases: PubMed/MEDLINE, Sabinet, EBSCOhost, CINAHL, Scopus and ScienceDirect, for relevant studies published between 2010-2020 globally, and grey literature. Identified articles will be exported into Mendeley Reference Management software and two reviewers will independently screen the titles, abstracts and full texts for eligibility. Any discrepancies between the two reviewers will be resolved by a third reviewer. Studies reporting on interventions to improve retention, adherence and viral load suppression for adolescents will be considered for inclusion. The systematic review will be performed and reported according to the Preferred Reporting Items for Systematic reviews and Meta-Analyses Protocols. Where feasible, a meta-analysis will be conducted using Stata Statistical Software: Release 16. The quality of the studies and risk of bias will be assessed using the CASP checklists and ROBINS-I tool, respectively.

Ethics and dissemination: The systematic review entails abstracting and reviewing already publicly available data rather than any involvement of participants, therefore no ethical clearance will be required. Results will be shared with relevant policy makers, program managers and service providers, and published and share through conferences and webinars.

PROSPERO registration number: CRD42021232440

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Strengths and limitations of this study

- This study contributes to the knowledge around available and evaluated interventions targeting adolescents with unsuppressed viral loads that could inform adolescent-friendly HIV services guidelines in both health facilities and community settings.
- This protocol utilizes the standardized Preferred Reporting Items for Systematic reviews and Meta-Analyses Protocols 2015 guidelines.
- The search strategy utilizes six major electronic databases which will include peer-reviewed articles and will include grey literature.
- A potential limitation of this study includes the limitation to studies published in English language only, which has the potential to miss other relevant studies published in other languages.
- The limitations of this study may be related to the timeline of the search strategy (2010 – 2020).

Introduction

Due to the successes in prevention of mother-to-child transmission (PMTCT) programs worldwide and advances in pediatric HIV treatment, children with HIV are surviving to reach adolescence (1). Innovations for Early Infant Diagnosis such as Point of Care diagnostics and prompt initiation on life-saving HIV treatment, has meant a large population of perinatally HIV-infected children are growing into adolescence (2). A mixed adolescent population exists consisting of perinatally infected and behaviorally infected adolescents living with HIV (ALHIV) who may respond to interventions differently (3). Worldwide, an estimated 1.6 million adolescents between 10-19 years were living with HIV in 2018; with nearly 85% living in sub-Saharan Africa (4)(5) . The World Health Organization (WHO) defines adolescents as children or young adults between 10-19 years of age (6). Although data on HIV treatment among adolescents is scarce, access and uptake of comprehensive care and treatment packages among this vulnerable population group is reportedly lower than adults (7). In 2019, only 53% (36-64%) of children aged 0-14 years were accessing treatment as compared to 68% (54-80%) coverage among 15 years and older individuals living with HIV (8). In most sub-Saharan Africa countries, public health facilities are ill-equipped to give guidance and support for ALHIV to remain engaged in HIV care and adhere to medication regimens (9).

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Good adherence to treatment regimens and consistent engagement in HIV care is vital for achieving sustained viral suppression, which is essential for ensuring the wellbeing of ALHIV and reducing chances of onward HIV transmission (10). Poor ART adherence increases the risk of viral drug-resistance, limits treatment efficacy, leading to disease progression, and reduces future therapeutic options as well as increasing the risk of transmission due to unsuppressed viral replication (11). Although reported ART adherence is high globally (>95%), concerns have been raised about waning adherence over time including loss of patients from HIV programs when scaling up (12). Evidence-based interventions to address adherence challenges for people on ART include individual and group adherence counselling, mHealth platforms, community and home-based strategies, pharmacist counselling and monitoring, task-shifting, medication fast-tracking, nutrition support and provision of disability grants (13). For ALHIV, individual counselling, group counselling and peer support, such as in teen clubs, have been some of the most common interventions.

Adolescents living with HIV have unique needs and are notably underserved globally and in national responses, which negatively affects their access to ART and results in poor ART adherence and inferior treatment outcomes such as achieving and maintain virologic suppression (14). According to WHO, a maintained viral load of <1000 ribonucleic acid (RNA) copies per ml of plasma is considered evident of virologic suppression (15). However recent debates have suggested reducing the suppression threshold to 200 copies/ml, informed by evidence from HPTN 052 findings, PARTNERS and Opposites Attract, who all reported that successful ART with viral suppression prevented transmission when viral load was undetectable below 200 copies/mL (16). Variations in viral load assay limit of detection has also been tied to the debates around threshold setting for viral load suppression, and these need to be paid attention to in any evaluation.

A 2014 systematic review showed global adherence estimates ranging from 16% to 99% among adolescents whilst a meta-analysis of data for young adults and adolescents aged 12–24 years in 53 countries found an estimated pooled adherence at 84% in Asia and Africa. Rates of viral suppression (< 400 copies/mL) ranged from 52% to 87% in Asia, 27% to 89% in Africa, and from 37.5% to 49% in Central and South America (17). A recent review of eleven studies done in sub-Saharan Africa identified a complex web of 44 barriers and 29 facilitators impacting ART

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adherence among adolescents, with stigma, lack of assistance, ART side-effects and forgetfulness as top barriers whilst top facilitators were caregiver support, knowledge of status and peer support groups (18). The barriers and facilitators of ART adherence among adolescents in low- and middle-income countries have been noted as family structure, a burdensome ART regimen, attitudes about medication and various healthcare system factors (19).

Despite recent increased focus and several systematic reviews synthesizing interventions to improve treatment outcomes among adolescents, we could not find reviews specifically focusing on studies targeting unsuppressed adolescents. Previous reviews have identified specific types of interventions that show promise for improved adherence to ART and retention in HIV care with very limited quantity of studies and low quality in several studies focusing on adolescents aged 10-19 years published up until 2015 (20). A review by MacPherson *et al*. focused on studies published between 2001 and 2014 evaluating effectiveness of interventions on service delivery to improve linkage to treatment initiation, retention in HIV care, and ART adherence among ALHIV aged 10-19 years (14). Considering that only 11 studies were included in the study, with low-to-moderate methodological quality, from high-income countries and none from countries with a generalized HIV epidemic, the authors suggested further investigations on their findings which included promising results from individual and group education and counseling, increased accessibility to clinics, youth-friendly services and financial incentives (14).

Casale *et al.* reviewed studies published between January 2016 and June 2018, and provided an update on evidence from empirical evaluations on interventions designed to enhance ART adherence and retention in care for adolescents aged 10-19 years and youth aged 15-24 years living with HIV (20). The authors reported on 10 relevant studies, mostly from sub-Saharan Africa, utilizing six clinic-level interventions, three community or household level intervention, and one mHealth trial at individual level, and they concluded that there is need to further develop and test multifaceted interventions beyond facility-based interventions, that address broader socio-economic barriers, prioritizing adolescents aged 10-19 years, if we are to address the treatment gap among this population group (20). A more recent review of articles published between September 2015 and January 2019 by Lindsey *et al.* reported that ART adherence interventions for adolescents and youth implemented in low- and middle-income countries showed mixed results

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from interventions at patient-level and health services level, with none of the interventions improving ART adherence or viral load suppression (21). The authors argued that intensive home-based case management models of care hold much promise in improving treatment outcomes in adolescents and youth and warrant further research (21). Most recent published studies focusing on interventions such as Enhanced Adherence Counseling (EAC) and Community Adolescent Treatment Supporters (CATS) to improve treatment outcomes among ALHIV challenged with viral suppression and retention in care may provide new evidence for consideration (22)(23)(24)

Rationale for the review

Globally, it is apparent that there has been an increased focus in addressing the treatment gap among ALHIV, and in recent years, there have been a number of systematic reviews focusing on interventions to improve ART adherence and retention in HIV care among adolescents and young adults in general. While the gaps described above and the systematic reviews cited present opportunities and specific interventions that show promise, they also highlight the limited quantity and low-to-moderate quality of studies and evidence on interventions targeting ALHIV. These reviews commonly include interventions for all adolescents and youth, despite their viral load suppression levels, which include clinic-level interventions, individual-level, and community- or household-level interventions. Considering the urgent need to design, implement, and test interventions to keep adolescents in care and achieve and maintain viral load suppression, we propose to review studies that specifically evaluated interventions targeting adolescents challenged with staying engaged in care and maintaining viral suppression. This review will focus on adolescents experiencing poor treatment outcomes to provide an updated data synthesis of all adherence and retention interventions to close the treatment gap. We aim to broaden the synthesized knowledge base to include new and innovative interventions for this particular population, at health facility, community and household levels, and potentially reinforce or modify already existing promising interventions for achievement of sustained viral suppression.

Methods

This systematic review will use the seven steps as described by Eggar, Davey and Smith (25). The seven steps are outlined as (1) formulate the review question; (2) define the inclusion and exclusion

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criteria; (3) develop a search strategy; (4) study selection; (5) assess the quality of studies; (6) extract data, and (7) analyse or synthesise the data.

Review question

The following questions will guide the review of the literature published between 2010 and 2020:

1. What interventions are being implemented to improve treatment outcomes for adolescents on ART?
2. How effective have the interventions been in achieving better treatment outcomes among adolescents on ART?

Inclusion and exclusion criteria

The literature search will be guided by the following inclusion criteria:

1. Published studies in peer-reviewed journals and grey literature
2. The study *population* includes ALHIV aged 10-19 years
3. Reported the association between the *intervention*, and retention, adherence and viral load suppression
4. Reported *comparisons* in quantitative measures of retention, adherence and viral load suppression, including pre-/post- or multi-arm comparison groups.
5. Study *outcomes* include retention, adherence and viral load suppression
6. Studies published in English language
7. Studies from all countries published between 2010 and 2020.

Studies that are not available in English language will be excluded, as well as review papers, editorials and letters.

The decision to only include studies reported in English language was necessitated by the limitations in capacity and resources among the review team, which restricts the review to English language only. Studies may include younger children below 10 years or young adults over 19 years, and justification for inclusion will be provided if these are included in the analysis. Additionally, other studies may focus on adherence interventions and viral load suppression for all people living with HIV, including adolescents in the study population or sample. Age

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disaggregation will be performed where there are age overlaps and the authors have provided age-disaggregated data.

Table 1. PICOT based inclusion criteria

Patient population	Adolescents aged 10-19 years living with HIV, acquired perinatally or behaviourally.
Interventions	Any group-based or individual-level interventions to improve treatment outcomes targeting adolescents who have challenges suppressing their viral loads, including Differentiated Service Delivery (DSD) models (at facility, community or homebased levels).
Comparisons	Studies may report comparisons on quantitative measures of retention, adherence and viral load suppression between intervention and standard care (pre-/post- or multi-arm comparison groups).
Outcomes	Primary outcome - viral load suppression Secondary outcomes – retention in care and ART adherence
Time	2010 – 2020
Other considerations	Language: English
	Study designs: Quantitative and Mixed methods
	Geographical area: Global

Outcome measures

ART adherence – adherence can be measured using pill counts, pharmacy refill data, self-reporting, as well as physician/nurse assessments and appointment keeping. More robust measures such as Medical Events Monitoring System (MEMS) and other new technologies will all be considered in this review. Optimal adherence will be considered to be at ≥ 95% of prescribed medication doses taken per specified period.

Retention in HIV care **–** methods of measuring retention in care may be based on missed clinic visits or based on kept clinic visits. Retention variables of interest may also include whether patient

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is alive and on continued ART, defaulted/lost, lost to follow-up (LTFU), transferred-out or died. A sub-analysis of the different categories of adolescents who are not still engaged in care will be performed.

Viral load suppression – although WHO viral load suppression threshold in 1000 copies/mL, more recent studies may have used lower levels of viral load counts to define viral load suppression or disaggregate low level viremia and suppression. All definitions of viral load suppression will be considered in this review. Where different cut-offs for viral load suppression are used, we will run sensitivity analyses.

Search strategy for identification of studies

The search strategy was developed with the assistance of the faculty Librarian who will provide further guidance and advice in conducting the searches for the review. A comprehensive database search of literature reporting on treatment interventions and viral load suppression will be systematically conducted in six major databases: PubMed/MEDLINE, Sabinet, EBSCOhost, CINAHL, Scopus and ScienceDirect. Grey literature searches will be performed on OpenGrey, Grey Literature report, International AIDS Conference abstract archive, and HIV and Adolescence. We will search the identified databases for all articles published between 2010 and 2020 that meet the specified selection criteria. The search syntax will first be used to perform a database search on PubMed/MEDLINE to test and optimize the search strategy, which will subsequently be replicated and adapted as needed for other database searches. Full-text articles will be sourced using a list of key words and terms for the search strategy using the "AND" and "OR" Boolean operators as appropriate. The search terms will include "Adolescents", "antiretroviral therapy", "retention", "adherence", "viral load", "suppression", "unsuppressed". The search strategies used for the selected databases are provided in Supplemental Material titled "ALL Database search Results".

Study selection

Studies will be included in the systematic review using the Population/Patient/Problem, Intervention, Comparison, Outcome and Time (PICOT) mnemonics. We will also consider studies done using quantitative or mixed methods, with a worldwide search without any particular

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geographical location limitations. Restricting reviews to a particular study design, such as RCTs only, may restrict findings to effectiveness only rather than seeking answers to more complex phenomena such as why an intervention may or may not be effective (26).

The selection, screening and reporting of this review will be done in accordance with the reporting guidance provided in the Preferred Reporting Items for Systematic Reviews and Meta-Analyses Protocol (PRISMA-P) 2015 guidelines (27). Selected studies will be collated and exported into Mendeley Reference Management Software, which will be used to manage the citations and recording the number of hits for each article from all databases, as well as removing duplicate records (28). Two reviewers will independently review titles and abstracts of all articles retrieved to assess eligibility for inclusion. The two reviewers will further review full-texts of the selected articles independently, to establish compliance with the eligibility criteria for inclusion in the final assessment (25). Any disagreements on inclusion or exclusion of any article will be referred to a third reviewer authorized to resolve any disagreements for inclusion of articles. A PRISMA flow diagram will be presented to describe the study selection decisions made.

Data Extraction & Quality assessment

Conducting a quality assessment may determine the inclusion or exclusion of an initially selected study, and this should consider the robustness (validity and reliability) of findings of a particular study (26). We will create a data extraction matrix in Microsoft Excel to capture study characteristics and outcomes data, such as Author(s), study year and duration, publication year, geographical location, target population, sample, purpose, study design, intervention, data collection and analyses, outcomes, results, limitations and conclusions. Only the data reported in the articles will be extracted and used in the quality assessment although additional information may be sought from authors for clarification.

Data analysis/synthesis: Assessment of evidence using the GRADE system

The two primary review authors will independently assess the selected articles using the GRADE (grading of recommendations assessment, development, and evaluation) considerations to downgrade or upgrade the certainty of evidence on the interventions and the specific outcomes reported by each study. Independent conclusions will be drawn on how the interventions compare

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to each other, on the specified treatment outcomes considering the estimates of effects, and the certainty of the evidence (risk of bias, confidence in evidence/ consistency in effects, quality of evidence, imprecision, indirectness, and publication bias) (29))(30). The quality of evidence and risk of bias for non-randomized control trials will be assessed using the Critical Appraisal Skills Programme (CASP) Checklist and the Risk of Bias in Non-randomized Studies of Interventions (ROBINS-I) Assessment Tool, respectively (31). Studies which included clinical or medication issues such as drug resistance, and not necessarily adherence and retention challenges only, may be considered with appropriate justifications, especially reports on recent improved ART regimens.

We will conduct a meta-analysis only if the interventions, research questions or focus, and study participants allow for a meaningful meta-analysis. The meta-analysis will be conducted using Stata Statistical Software: Release 16. College Station, TX: StataCorp LLC. The X^2 test and the Higgins I^2 statistic will be used to quantify statistical heterogeneity by estimating the variability between study estimates, with I^2 statistic values of 25%, 50%, and 75% representing low, medium and high heterogeneity respectively (32). Forest plots will be used to visualize the extent of heterogeneity or homogeneity among the selected studies.

Publication bias described as the tendency by authors to report positive findings, will be presented and visualized on a funnel plot, as well as statistically assessed using the Harbord`s regression test to assess the asymmetry of the funnel plot (33)) ((34). The contour-enhanced funnel plot will be used to distinguish between asymmetry caused by publication bias versus other causes, which may not have been successfully accomplished by the standard method (35). If it is not possible to do a meta-analysis, the findings extracted onto the Excel spreadsheet described above will be summarized and reported in a table form presenting key components of the studies. Subgroup analysis may include facility vs community-based interventions and younger vs older adolescents.

Patient and Public involvement

No patients or public population is involved in the development and conducting of this study, as this is a protocol for a systematic review of already available literature.

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Ethics and dissemination

The systematic review entails abstracting and reviewing already publicly available data rather than any involvement of participants, therefore no ethical clearance or approval will be required.

We plan to share results of this review with relevant policy makers and guidelines developers, program managers and service providers, particularly providing evidence-based guidance for interventions targeting unsuppressed adolescents. We also plan to publish the results of the review, as well as sharing the findings with key stakeholders using platforms such as conferences, and presentations through webinars and the HIV Project ECHO (Extension for Community Health Outcomes) (36).

Contributions: FKM conceived the idea, developed the research questions and study methods and contributed meaningfully to the drafting and editing, he also approved the final manuscript. BvW aided in developing the research questions and study methods, contributed meaningfully to the drafting and editing, and approved the final manuscript.

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Competing interests: None declared

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References

1. Bekker L-G, Siberry GK, Hirnschall G. Ensuring Children and Adolescents Are Not Left Behind [Internet]. 2018 [cited 2020 May 6]. Available from: https://journals.lww.com/jaids

2. UNICEF. HIV and AIDS in Adolescents - UNICEF Data [Internet]. 2020 [cited 2020 Dec 11]. Available from: https://data.unicef.org/topic/adolescents/hiv-aids/

3. Sherr L, Cluver LD, Toska E, He E. Differing psychological vulnerabilities among behaviourally and perinatally HIV infected adolescents in South Africa–implications for targeted health service provision*. AIDS Care - Psychol Socio-Medical Asp AIDS/HIV [Internet]. 2018 Jun 20 [cited 2021 May 25];30:92–101. Available from: https://www.tandfonline.com/doi/abs/10.1080/09540121.2018.1476664

4. UNICEF. Adolescent HIV prevention - UNICEF DATA [Internet]. 2020 [cited 2020 Sep 10]. Available from: https://data.unicef.org/topic/hivaids/adolescents-young-people/

5. UNAIDS. UNAIDS data 2019 [Internet]. 2019 [cited 2020 May 6]. Available from: https://www.unaids.org/sites/default/files/media_asset/2019-UNAIDS-data_en.pdf

6. WHO. WHO | Definition of key terms. WHO [Internet]. 2013 [cited 2016 Oct 11]; Available from: http://www.who.int/hiv/pub/guidelines/arv2013/intro/keyterms/en/

7. UNAIDS. Ending the AIDS epidemic for adolescents, with adolescents: A practical guide to meaningfully engage adolescents in the AIDS response. Geneva; 2016.

8. UNAIDS. Global HIV & AIDS statistics — 2020 fact sheet | UNAIDS [Internet]. 2020 [cited 2020 Dec 12]. Available from: https://www.unaids.org/en/resources/fact-sheet

9. Ssali L, Kalibala S, Birungi J, Egessa A, Wangisi J, Lyavala J, et al. Retention of adolescents living with HIV in care, treatment, and support programs in Uganda. 2014 [cited 2018 Feb 27]; Available from: http://www.hivcore.org/Pubs/Uganda_AdolHAART_Rprt.pdf

10. Grimsrud AT, Pike C, Bekker LG. The power of peers and community in the continuum

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

of HIV care [Internet]. Vol. 8, The Lancet Global Health. Elsevier Ltd; 2020 [cited 2020 Nov 16]. p. e167–8. Available from: https://pubmed.ncbi.nlm.nih.gov/31924538/

11. Kim S-H, Gerver SM, Fidler S, Ward H. Adherence to antiretroviral therapy in adolescents living with HIV: systematic review and meta-analysis. AIDS [Internet]. 2014 Aug 24 [cited 2018 Feb 27];28(13):1945–56. Available from: http://www.ncbi.nlm.nih.gov/pubmed/24845154

12. Nachega JB, Mills EJ, Schechter M. Antiretroviral therapy adherence and retention in care in middle-income and low-income countries: current status of knowledge and research priorities. Curr Opin HIV AIDS [Internet]. 2010 Jan [cited 2018 Sep 30];5(1):70–7. Available from: https://insights.ovid.com/crossref?an=01222929-201001000-00011

13. Ridgeway K, Dulli LS, Murray KR, Silverstein H, Santo LD, Olsen P, et al. Interventions to improve antiretroviral therapy adherence among adolescents in low- and middle-income countries: A systematic review of the literature [Internet]. Vol. 13, PLoS ONE. 2018. 1–33 p. Available from: http://dx.doi.org/10.1371/journal.pone.0189770

14. MacPherson P, Munthali C, Ferguson J, Armstrong A, Kranzer K, Ferrand RA, et al. Service delivery interventions to improve adolescents' linkage, retention and adherence to antiretroviral therapy and HIV care. Trop Med Int Health [Internet]. 2015 Aug [cited 2016 Sep 30];20(8):1015–32. Available from: http://www.ncbi.nlm.nih.gov/pubmed/25877007

15. WHO. WHO | Viral suppression after 12 months of antiretroviral therapy in low- and middle-income countries: a systematic review. WHO [Internet]. 2013 [cited 2017 Jul 13]; Available from: http://www.who.int/bulletin/volumes/91/5/12-112946/en/

16. WHO. WHO | Viral suppression for HIV treatment success and prevention of sexual transmission of HIV [Internet]. 2018 [cited 2020 Dec 12]. Available from: https://www.who.int/hiv/mediacentre/news/viral-supression-hiv-transmission/en/

17. Ridgeway K, Dulli LS, Murray KR, Silverstein H, Santo LD, Olsen P, et al. Interventions to improve antiretroviral therapy adherence among adolescents in low- and middle-income countries: A systematic review of the literature [Internet]. Vol. 13, PLoS ONE. Public Library of Science; 2018 [cited 2020 Jul 9]. Available from:

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

/pmc/articles/PMC5749726/?report=abstract

18. Ammon N, Mason S, Corkery JM. Factors impacting antiretroviral therapy adherence among human immunodeficiency virus-positive adolescents in Sub-Saharan Africa: a systematic review. Public Health [Internet]. 2018 Apr 1 [cited 2018 Sep 30];157:20–31. Available from: http://www.ncbi.nlm.nih.gov/pubmed/29501984

19. Hudelson C, Cluver L. Factors associated with adherence to antiretroviral therapy among adolescents living with HIV/AIDS in low- and middle-income countries: a systematic review. AIDS Care [Internet]. 2015 [cited 2016 Sep 30];27(7):805–16. Available from: http://www.ncbi.nlm.nih.gov/pubmed/25702789

20. Casale M, Carlqvist A, Cluver L. Recent Interventions to Improve Retention in HIV Care and Adherence to Antiretroviral Treatment Among Adolescents and Youth: A Systematic Review. AIDS Patient Care STDS [Internet]. 2019 Jun 1 [cited 2020 Apr 26];33(6):237–52. Available from: https://www.liebertpub.com/doi/10.1089/apc.2018.0320

21. Reif LK, Abrams EJ, Arpadi S, Elul B, McNairy ML, Fitzgerald DW, et al. Interventions to Improve Antiretroviral Therapy Adherence Among Adolescents and Youth in Low- and Middle-Income Countries: A Systematic Review 2015–2019 [Internet]. Vol. 1, AIDS and Behavior. Springer; 2020 [cited 2020 Jul 31]. p. 1. Available from: /pmc/articles/PMC7223708/?report=abstract

22. Mavhu W, Willis N, Mufuka J, Bernays S, Tshuma M, Mangenah C, et al. Effect of a differentiated service delivery model on virological failure in adolescents with HIV in Zimbabwe (Zvandiri): a cluster-randomised controlled trial. Lancet Glob Heal [Internet]. 2020 Feb 1 [cited 2020 Apr 26];8(2):e264–75. Available from: https://www.sciencedirect.com/science/article/pii/S2214109X19305261

23. Diress G, Dagne S, Alemnew B, Adane S, Addisu A. Viral Load Suppression after Enhanced Adherence Counseling and Its Predictors among High Viral Load HIV Seropositive People in North Wollo Zone Public Hospitals, Northeast Ethiopia, 2019: Retrospective Cohort Study. AIDS Res Treat. 2020;2020.

24. Nasuuna E, Kigozi J, Babirye L, Muganzi A, Sewankambo NK, Nakanjako D. Low HIV

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

viral suppression rates following the intensive adherence counseling (IAC) program for children and adolescents with viral failure in public health facilities in Uganda. BMC Public Health [Internet]. 2018 Dec 22 [cited 2021 Mar 27];18(1):1048. Available from: https://bmcpublichealth.biomedcentral.com/articles/10.1186/s12889-018-5964-x

25. Egger M, Smith GD. Principles of and Procedures for Systematic Reviews. In: Systematic Reviews in Health Care: Meta-Analysis in Context: Second Edition. Wiley Blackwell; 2008. p. 23–42.

26. Grant MJ, Booth A. A typology of reviews: An analysis of 14 review types and associated methodologies [Internet]. Vol. 26, Health Information and Libraries Journal. John Wiley & Sons, Ltd; 2009 [cited 2020 Nov 3]. p. 91–108. Available from: https://onlinelibrary.wiley.com/doi/full/10.1111/j.1471-1842.2009.00848.x

27. Moher D, Shamseer L, Clarke M, Ghersi D, Liberati A, Petticrew M, et al. Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015 statement. Rev Esp Nutr Humana y Diet [Internet]. 2016 [cited 2020 Nov 3];20(2):148–60. Available from: https://pubmed.ncbi.nlm.nih.gov/25554246/

28. Price A, Price, Amy. Mendeley and More for Systematic Reviews. Int Netw Knowl about Wellbeing [Internet]. 2016 Apr 26 [cited 2020 Nov 3]; Available from: http://www.ithinkwell.org/mendeley-and-more-for-systematic-reviews/

29. Mearns H, Otiku PK, Shelton M, Kredo T, Kagina BM, Schmidt BM. Screening strategies for adults with type 2 diabetes mellitus: A systematic review protocol [Internet]. Vol. 9, Systematic Reviews. BioMed Central; 2020 [cited 2020 Nov 16]. p. 156. Available from: https://systematicreviewsjournal.biomedcentral.com/articles/10.1186/s13643-020-01417-3

30. Brignardello-Petersen R, Izcovich A, Rochwerg B, Florez ID, Hazlewood G, Alhazanni W, et al. GRADE approach to drawing conclusions from a network meta-analysis using a partially contextualised framework. BMJ [Internet]. 2020 Nov 10 [cited 2020 Nov 16];371:m3907. Available from: https://www.bmj.com/lookup/doi/10.1136/bmj.m3907

31. Haghighat R, Steinert J, Cluver L. The effects of decentralising antiretroviral therapy care delivery on health outcomes for adolescents and young adults in low- and middle-income

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

countries: a systematic review. Glob Health Action [Internet]. 2019 Jan 1 [cited 2021 Mar 15];12(1):1668596. Available from: https://www.tandfonline.com/doi/full/10.1080/16549716.2019.1668596

32. Mengesha MM, Ajema D, Teshome A, Tura AK. The association between diagnosis disclosure and adherence to antiretroviral therapy among adolescents living with HIV in sub-Saharan Africa: A protocol for systematic review and meta-analysis [Internet]. Vol. 9, Systematic Reviews. BioMed Central; 2020 [cited 2020 Nov 7]. p. 160. Available from: https://systematicreviewsjournal.biomedcentral.com/articles/10.1186/s13643-020-01420-8

33. Egger M, Smith GD, Schneider M, Minder C. Bias in meta-analysis detected by a simple, graphical test. Br Med J [Internet]. 1997 Sep 13 [cited 2020 Nov 16];315(7109):629–34. Available from: https://www.bmj.com/content/315/7109/629

34. Rao G, Lopez-Jimenez F, Boyd J, D'Amico F, Durant NH, Hlatky MA, et al. Methodological standards for meta-analyses and qualitative systematic reviews of cardiac prevention and treatment studies a scientific statement from the American Heart Association. Circulation [Internet]. 2017 Sep 1 [cited 2020 Nov 16];136(10):e172–94. Available from: https://pubmed.ncbi.nlm.nih.gov/28784624/

35. Chaimani A, Mavridis D, Salanti G. A hands-on practical tutorial on performing meta-analysis with Stata. Evid Based Ment Health [Internet]. 2014 Nov 1 [cited 2020 Nov 16];17(4):111–6. Available from: https://pubmed.ncbi.nlm.nih.gov/25288685/

36. Bikinesi L, O'Bryan G, Roscoe C, Mekonen T, Shoopala N, Mengistu AT, et al. Implementation and evaluation of a Project ECHO telementoring program for the Namibian HIV workforce. Hum Resour Health [Internet]. 2020 Sep 1 [cited 2020 Nov 17];18(1):1–10. Available from: https://link.springer.com/articles/10.1186/s12960-020-00503-w

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PubMed Results – 23.02.2021 = 64 Results

(adolescents or teenagers or young adults or teen or youth or student or adolescence) AND (antiretroviral therapy) AND (retention) AND (adherence or compliance) AND (viral load) AND (suppression) AND (hiv or aids or acquired human immunodeficiency syndrome or human immunodeficiency virus)

Filters applied: Humans, English, Adolescent: 13-18 years, Adult: 19+ years, Child: 6-12 years, from 2010 - 2020.

Sabinet Results – 23.02.2021 = 68 Results

```
[[All: adolescents] OR [All: teenagers] OR [All: young]] AND [[All: adults]
OR [All: teen] OR [All: youth] OR [All: student]] AND [All: or adolescence]
AND [All: antiretroviral] AND [[[All: therapy] AND [All: retention] AND
[All: adherence]] OR [[All: compliance] AND [All: viral]]] AND [All: load]
AND [All: suppression] AND [Languages: English] AND [Publication Date:
(01/01/2010 TO 12/31/2020)]
```

ScienceDirect – 08.03.2021 = 834 Results

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adolescents AND antiretroviral therapy AND retention AND adherence OR
compliance AND viral load AND suppression AND hiv or aids AND 10-19 years,
Filters applied: 2010 - 2020
```

Scopus – 09.03.2021 = 59 Results

TITLE-ABS-KEY ((adolescents OR teenagers OR young AND adults OR teen OR youth OR student OR adolescence) AND (antiretroviral AND therapy) AND (retention) AND (adherence OR compliance) AND (viral AND load) AND (suppression) AND (hiv OR aids OR acquired AND human AND immunodeficiency AND syndrome OR human AND immunodeficiency AND virus)) Additional Filter: 2010 – 2020

Web of Science – 09.03.2021 = 37 Results

((adolescents or teenagers or young adults or teen or youth or student or adolescence) AND (antiretroviral therapy) AND (retention) AND (adherence or compliance) AND (viral load) AND (suppression) AND (hiv or aids or acquired human immunodeficiency syndrome or human immunodeficiency virus))
Timespan: 2010-2020. **Indexes:** SCI-EXPANDED, SSCI, A&HCI, CPCI-S, CPCI-SSH, BKCI-S, BKCI-SSH, ESCI.

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EBSCOhost – 15.03.2021 = 573 Results

"adolescents or teenagers or young adults or teen or youth AND antiretroviral therapy or antiretroviral treatment AND adherence AND retention AND viral load AND HIV Published Date: 20100101-20201231 AND Apply equivalent subjects on 2021-03-15 10:19 AM"

adolescents or teenagers or young adults or teen or youth AND antiretroviral therapy or antiretroviral treatment AND adherence AND retention AND viral load AND HIV
https://search-ebscohost-com.ezproxy.uwc.ac.za/login.aspx?direct=true&db=rzh&bquery=adolescents+or+teenagers+or+young+adults+or+teen+or+youth+AND+antiretroviral+therapy+or+antiretroviral+treatment+AND+adherence+AND+retention+AND+viral+load+AND+HIV&cli0=DT1&clv0=201001-202012&type=1&searchMode=Relevancy&site=ehost-live&ssl=y

Limiters - Published Date: 20100101-20201231
Expanders - Apply equivalent subjects
Narrow by Language: - english
Narrow by SubjectAge: - child: 6-12 years
Narrow by SubjectAge: - adolescent: 13-18 years
Search modes - SmartText Searchin

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Interventions for improving treatment outcomes in Adolescents on antiretroviral therapy with unsuppressed viral loads: a systematic review protocol

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Abstract

Introduction: Adolescents represent one of the most underserved population groups among people living with HIV. With successes in the elimination of Mother to Child transmission initiatives and advances in pediatric HIV treatment programs, a large population of HIV-infected children are surviving into adolescence. Adolescence presents unique challenges that increases the risk of non-suppressed viral loads in adolescents living with HIV (ALHIV). There is need to develop, implement and test interventions to improve viral suppression among ALHIV. Systematic reviews of recent studies present scarce and inconclusive evidence of effectiveness of current interventions, especially for adolescents. This protocol provides a description of a planned review of interventions to improve treatment outcomes among unsuppressed ALHIV.

Methods and analysis: A comprehensive search string will be used to search six bibliographic databases: PubMed/MEDLINE, Sabinet, EBSCOhost, CINAHL, Scopus and ScienceDirect, for relevant studies published between 2010-2020 globally, and grey literature. Identified articles will be exported into Mendeley Reference Management software and two independent reviewers will screen the titles, abstracts and full texts for eligibility. A third reviewer will resolve any discrepancies between the two initial reviewers. Studies reporting on interventions to improve viral suppression, retention and adherence for adolescents will be considered for inclusion. The systematic review will be performed and reported according to the Preferred Reporting Items for Systematic reviews and Meta-Analyses Protocols. Where feasible, a meta-analysis will be conducted using Stata Statistical Software: Release 16. The quality of the studies and risk of bias will be assessed using the CASP checklists and ROBINS-I tool, respectively.

Ethics and dissemination: The systematic review entails abstracting and reviewing already publicly available data rather than any involvement of participants, therefore no ethical clearance will be required. Results will be shared with relevant policy makers, program managers and service providers, and published and share through conferences and webinars.

PROSPERO registration number: CRD42021232440

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Strengths and limitations of this study

- This review contributes to the available knowledge on evaluated interventions targeting adolescents with unsuppressed viral loads that could inform adolescent-friendly HIV services guidelines in both health facilities and community settings.
- This protocol utilizes the standardized Preferred Reporting Items for Systematic reviews and Meta-Analyses Protocols 2015 guidelines.
- The search strategy utilizes six major electronic databases which will include peer-reviewed articles and grey literature.
- A potential limitation of this study includes the limitation to studies published in English language only, which has the potential to miss other relevant studies published in other languages.
- The study limitations may be associated with the timeline for the search strategy (2010 – 2020).

Introduction

Along with the successes in elimination of mother-to-child transmission (*e*MTCT) initiatives worldwide and developments in pediatric HIV treatment, more children with HIV are surviving to reach adolescence (1). Innovations for Early Infant Diagnosis such as Point of Care diagnostics and prompt initiation of life-saving HIV treatment, has meant a large population of perinatally HIV-infected children are growing into adolescence (2). A mixed adolescent population exists consisting of perinatally infected and behaviorally infected adolescents living with HIV (ALHIV) who may respond to interventions differently (3). Worldwide, 1.6 million adolescents aged 10-19 years were estimated to be living with HIV in 2018; with approximately 85% residing in sub-Saharan Africa (4)(5) . The World Health Organization (WHO) defines adolescents as individuals aged between 10-19 years (6). The proposed systematic review seeks to look at literature on both perinatally and behaviorally infected adolescents, aged 10-19 years. Although data on HIV treatment among adolescents is scarce, access and uptake of comprehensive care and treatment packages among this vulnerable population group is reportedly lower than adults (7). In 2019, only 53% (36-64%) of children aged 0-14 years were accessing treatment as compared to 68% (54-80%) coverage among 15 years and older individuals living with HIV (8). In the majority of sub-

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Saharan Africa countries, public sector health settings are ill-equipped to provide guidance and adequate support for ALHIV to stay engaged in HIV care and adhere to treatment regimens (9).

Good adherence to treatment regimens and consistent engagement in HIV care is essential for achieving sustained viral suppression, which is vital for ensuring the well-being of ALHIV and reducing chances of onward HIV transmission (10). Poor adherence to antiretrovirals increases the risk of emergence of drug resistance mutations, reduces treatment efficacy, thereby advancing disease progression, and reducing future treatment options and increasing the risk of further transmission driven by unsuppressed viral loads (11). Despite reportedly high levels of adherence to ART globally (>95%), diminishing adherence levels over time remains a concern, primarily the loss of patients during scaling up of HIV programs (12). Evidence-based strategies to mitigate adherence challenges for individuals on HIV treatment include facility and community-based interventions, which includes individual or group adherence counselling, pharmacist counselling and medication fast-tracking, mHealth, home-based/community strategies, financial incentives, nutritional support, motivational interviewing, and awarding of disability grants (13). Individual counselling, peer-led support groups and different models of group adherence counselling interventions, such as teen clubs, are most common interventions for ALHIV.

Adolescents receiving HIV treatment have distinct and unique needs and are notably underserved worldwide in national HIV responses. Ultimately, this negatively affects access to ART for adolescents, and results in suboptimal ART adherence and poorer treatment outcomes including achieving sustained viral suppression (14). The WHO defines viral suppression as a maintained viral load count of less than 1000 ribonucleic acid (RNA) copies/ml of plasma (15). However recent debates have suggested reducing the suppression threshold to 200 copies/ml, informed by evidence from HPTN 052 findings, PARTNERS and Opposites Attract, who all reported that successful ART with viral suppression prevented transmission when viral load was undetectable below 200 copies/mL (16). Variations in viral load assay limit of detection has also been tied to the debates around threshold setting for viral load suppression, and these need to be paid attention to in any evaluation.

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A systematic review conducted in 2014 estimated global ART adherence levels ranging between 16% - 99% amongst adolescents and a meta-analysis of data representing adolescents and young adults between 12–24 years found a pooled adherence estimate of 84% in 53 Asian and African countries. Viral suppression (cut-off < 400 copies/mL) rates were between 52% - 87% in Asia, 27% - 89% in Africa, and between 37.5% - 49% in Central and South America (17). A 2018 systematic review of 11 studies on factors impacting ART adherence among HIV positive adolescents in sub-Saharan Africa reported a complex web of 29 facilitators and 44 barriers among adolescents. These factors include stigma, forgetfulness, ART side-effects and lack of assistance as main barriers, whilst knowledge of status, caregiver support and peer support groups were the dominant facilitators (18). Other barriers and facilitators of treatment adherence among adolescents in countries in the global south have been noted as family support, a heavy pill burden/treatment regimen, attitudes towards medication and various health delivery system factors (19). Often these studies reported on adherence only using self-report or pill counts, with no viral load results reported. Self-reported adherence is mostly challenging, especially in adolescents, as clearly demonstrated in PrEP studies where there was no correlation between therapeutic drug monitoring and self-report especially in younger populations (20).

Despite recent increased focus and several systematic reviews synthesizing interventions to improve treatment outcomes among adolescents, we could not find reviews specifically focusing on studies targeting unsuppressed adolescents. Although our focus is predominantly on viral suppression, broader outcomes such as adherence and retention in care will be considered. Previous reviews have identified specific types of interventions that show promise for improved adherence to ART and retention in HIV care with very limited quantity of studies and low quality in several studies focusing on adolescents aged 10-19 years published up until 2015 (21). A review by MacPherson *et al*. focused on studies published between 2001 and 2014 evaluating effectiveness of interventions on service delivery to improve linkage to treatment initiation, retention in HIV care, and ART adherence among ALHIV aged 10-19 years (14). Considering that only 11 studies were included in the study, with low-to-moderate methodological quality, from high-income countries and none from countries with a generalized HIV epidemic, the authors suggested further investigations on their findings which included promising results from individual

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and group education and counseling, increased accessibility to clinics, youth-friendly services and financial incentives (14).

Casale *et al.* reviewed studies published between January 2016 and June 2018, and provided an update on evidence from empirical evaluations on interventions designed to enhance ART adherence and retention in care for adolescents aged 10-19 years and youth aged 15-24 years living with HIV (21). The authors reported on 10 relevant studies, mostly from sub-Saharan Africa, utilizing six clinic-level interventions, three community or household level intervention, and one mHealth trial at individual level, and they concluded that there is need for continued development and testing of multifaceted interventions over and above facility-based interventions, that address broader socio-economic barriers, prioritizing adolescents aged 10-19 years, if we are to address the treatment gap among this population group (21). A more recent review of studies published from September 2015 to January 2019 by Lindsey *et al.* reported that ART adherence interventions for adolescents and youth implemented in low- and middle-income countries showed mixed results from interventions at patient-level and health services level, with none of the interventions improving treatment adherence or virologic suppression (22). The authors argued that intensive home-based case management approaches to care hold much promise in improving treatment outcomes in adolescents and youth and warrant further research (22). Most recent published studies focusing on interventions such as Enhanced Adherence Counseling (EAC) and Community Adolescent Treatment Supporters (CATS) to improve treatment outcomes among ALHIV challenged with viral suppression and retention in care may provide new evidence for consideration (23)(24)(25)

Rationale for the review

Globally, it is apparent that there has been an increased focus in addressing the treatment gap among ALHIV, and in recent years, there have been a number of systematic reviews focusing on interventions to improve ART adherence and retention in HIV care among adolescents and young adults in general. While the gaps described above and the systematic reviews cited present opportunities and specific interventions that show promise, they also highlight the limited quantity and low-to-moderate quality of studies and evidence on interventions targeting ALHIV. These reviews commonly include interventions for all adolescents and youth, despite their viral load

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suppression levels, which include clinic-level interventions, individual-level, and community- or household-level interventions. Considering the urgent need to design, implement, and test interventions to keep adolescents in care and achieve and maintain viral load suppression, we propose to review studies that specifically evaluated interventions targeting adolescents challenged with staying engaged in care and maintaining viral suppression. This review will focus on adolescents experiencing poor treatment outcomes to provide an updated data synthesis of all adherence and retention interventions to close the treatment gap. We aim to broaden the synthesized knowledge base to include new and innovative interventions for this particular population, at health facility, community and household levels, and potentially reinforce or modify already existing promising interventions for achievement of sustained viral suppression.

Methods

This systematic review will use the seven steps as set out by Eggar, Davey and Smith (26). The seven steps are outlined as (1) formulate the review question; (2) define the inclusion and exclusion criteria; (3) develop a search strategy; (4) study selection; (5) assess the quality of studies; (6) extract data, and (7) analyse or synthesise the data.

Review question

The following questions will guide the review of the literature published between 2010 and 2020:

1. What interventions are being implemented to improve treatment outcomes for adolescents on ART with challenges suppressing their viral load?
2. How effective have the interventions been in achieving better treatment outcomes among adolescents on ART with challenges suppressing their viral load?

Inclusion and exclusion criteria

The literature search will be guided by the following inclusion criteria:

1. Published studies in peer-reviewed journals and grey literature
2. The study *population* includes ALHIV aged 10-19 years, both perinatally and behaviourally infected adolescents included.
3. Reported the association between the *intervention*, and viral suppression, retention and adherence

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4. Reported *comparisons* in quantitative measures of viral suppression, retention and adherence, including pre-/post- or multi-arm comparison groups.
5. Study *outcomes* include viral suppression, retention and ART adherence
6. Studies published in English language
7. Studies from all countries, high, middle- and low-income settings, published between 2010 and 2020.

The PICOT based inclusion criteria has been summarised in Table 1. Studies that are not available in English language will be excluded, as well as review papers, editorials and letters.

The decision to only include studies reported in English language was necessitated by the limitations in capacity and resources among the review team, which restricts the review to English language only. Studies may include younger children below 10 years or young adults over 19 years, and justification for inclusion will be provided if these are included in the analysis. Additionally, other studies may focus on adherence interventions and viral load suppression for all people living with HIV, including adolescents in the study population or sample. Age disaggregation will be performed where there are age overlaps and the authors have provided age-disaggregated data.

Table 1. PICOT based inclusion criteria

Patient population	Adolescents aged 10-19 years living with HIV, acquired perinatally or behaviourally, with challenges suppressing their viral load.
Interventions	Any group-based or individual-level interventions to improve treatment outcomes targeting adolescents who have challenges suppressing their viral loads, including Differentiated Service Delivery (DSD) models (at facility, community or homebased levels).
Comparisons	Studies may report comparisons on quantitative measures of retention, adherence and viral load suppression between

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	intervention and standard care (pre-/post- or multi-arm comparison groups).
Outcomes	Primary outcome - viral load suppression Secondary outcomes – retention in care and ART adherence
Time	2010 – 2020
Other considerations	Language: English
	Study designs: Quantitative and Mixed methods
	Geographical area: Global

Outcome measures

Viral load suppression – although WHO viral load suppression threshold in 1000 copies/mL, more recent studies may have used lower levels of viral load counts to define viral load suppression or disaggregate low level viremia and suppression. All definitions of viral load suppression will be considered in this review. Where different cut-offs for viral load suppression are used, we will run sensitivity analyses.

ART adherence – adherence can be measured using pill counts, pharmacy refill data, self-reporting, as well as physician/nurse assessments and appointment keeping. Self-report is a common measure of adherence although there are concerns about its validity (limitation in correlation to viral loads), and socially desirable responses from especially younger individuals may overestimate adherence (27). More robust measures such as Medical Events Monitoring System (MEMS) and other new technologies will all be considered in this review. Optimal adherence will be considered to be at ≥ 95% of prescribed medication doses taken per specified period.

Retention in HIV care – approaches to evaluating retention in HIV care may be determined from missed clinic appointments or based on kept clinic visits. Retention variables may also include interruption in treatment, lost to follow-up (LTFU), transferred-out or died. A sub-analysis of the different categories of adolescents who are not still engaged in care will be performed if possible (when different retention variables are reported).

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Search strategy for identification of studies

The search strategy was developed with the assistance of the faculty Librarian who will provide further guidance and advice in conducting the searches for the review. A comprehensive database search of literature reporting on treatment interventions and viral load suppression will be systematically conducted in six major databases: PubMed/MEDLINE, Sabinet, EBSCOhost, CINAHL, Scopus and ScienceDirect. Grey literature searches will be performed on OpenGrey, Grey Literature report, International AIDS Conference abstract archive, and HIV and Adolescence. We will search the identified databases for all articles published between 2010 and 2020 that meet the specified selection criteria. The search syntax will first be used to perform a database search on PubMed/MEDLINE to test and optimize the strategy, which will subsequently be replicated and adapted as needed for the other database searches. Full-text articles will be sourced using a list of key words and terms for the search strategy using the "AND" and "OR" Boolean operators as appropriate. The search terms will include "Adolescents", "antiretroviral therapy", "retention", "adherence", "viral load", "suppression", "unsuppressed". The search strategies used for the selected databases are provided in Supplemental Material titled "ALL Database search Results".

Study selection

Studies will be included in this systematic review according to the Population/Patient/Problem, Intervention, Comparison, Outcome and Time (PICOT) mnemonics. We will also consider studies done using quantitative or mixed methods, with a worldwide search without any particular geographical location limitations. Restricting reviews to a particular study design, such as RCTs only, may restrict findings to effectiveness only rather than seeking answers to more complex phenomena such as why an intervention may or may not be effective (28).

The selection, screening and reporting of this review will be done in accordance with the reporting guidance provided in the Preferred Reporting Items for Systematic Reviews and Meta-Analyses Protocol (PRISMA-P) 2015 guidelines (29). Selected studies will be collated and exported into Mendeley Reference Management Software, which will be used to manage the citations and recording the number of hits for each article from all databases, as well as removing duplicate records (30). Two reviewers will independently review titles and abstracts of all articles retrieved

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to assess eligibility for inclusion. The two reviewers will further review full-texts of the selected articles independently, to establish compliance with the eligibility criteria for inclusion in the final assessment (26). Any disagreements on inclusion or exclusion of any article will be referred to a third reviewer authorized to resolve any disagreements for inclusion of articles. A PRISMA flow diagram will be presented to describe the study selection decisions made.

Data Extraction & Quality assessment

Conducting a quality assessment may determine the inclusion or exclusion of an initially selected study, and this should consider the robustness (validity and reliability) of findings of a particular study (28). We will create a data extraction matrix in Microsoft Excel to capture study characteristics and outcomes data, such as Author(s), study year and duration, publication year, geographical location, target population, sample, purpose, study design, intervention, data collection and analyses, outcomes, results, limitations and conclusions. Only the data reported in the articles will be extracted and used in the quality assessment although additional information may be sought from authors for clarification.

Data analysis/synthesis: Assessment of evidence using the GRADE system

The two primary review authors will independently assess the selected articles using the GRADE (grading of recommendations assessment, development, and evaluation) considerations to downgrade or upgrade the certainty of evidence on the interventions and the specific outcomes reported by each study. Independent conclusions will be drawn on how the interventions compare to each other, on the specified treatment outcomes considering the estimates of effects, and the certainty of the evidence (risk of bias, confidence in evidence/ consistency in effects, quality of evidence, imprecision, indirectness, and publication bias) (31))(32). The quality of evidence and risk of bias for non-randomized control trials will be assessed using the Critical Appraisal Skills Programme (CASP) Checklist and the Risk of Bias in Non-randomized Studies of Interventions (ROBINS-I) Assessment Tool, respectively (33). Studies which included clinical or medication issues such as drug resistance, and not necessarily adherence and retention challenges only, may be considered with appropriate justifications, especially reports on recent improved ART regimens.

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We will conduct a meta-analysis only if the interventions, research questions or focus, and study participants allow for a meaningful meta-analysis. If possible, we will analyse results from the review by mode of infection among the adolescents included in meta-analysis. The meta-analysis will be conducted using Stata Statistical Software: Release 16. College Station, TX: StataCorp LLC. The X^2 test and the Higgins I^2 statistic will be used to quantify statistical heterogeneity by estimating the variability between study estimates, with I^2 statistic values of 25%, 50%, and 75% representing low, medium and high heterogeneity respectively (34). Forest plots will be used to visualize the extent of heterogeneity or homogeneity among the selected studies.

Publication bias described as the tendency by authors to report positive findings, will be presented and visualized on a funnel plot, as well as statistically assessed using the Harbord`s regression test to assess the asymmetry of the funnel plot (35)) ((36). The contour-enhanced funnel plot will be used to distinguish between asymmetry caused by publication bias versus other causes, which may not have been successfully accomplished by the standard method (37). If it is not possible to do a meta-analysis, the findings extracted onto the Excel spreadsheet described above will be summarized and reported in a table form presenting key components of the studies. Subgroup analysis may include facility vs community-based interventions and younger vs older adolescents.

Patient and Public involvement

No patients or public population is involved in the development and conducting of this study, as this is a protocol for a systematic review of already available literature.

Ethics and dissemination

The systematic review entails abstracting and reviewing already publicly available data rather than any involvement of participants, therefore no ethical clearance or approval will be required.

We plan to share results of this review with relevant policy makers and guidelines developers, program managers and service providers, particularly providing evidence-based guidance for interventions targeting unsuppressed adolescents. We also plan to publish the results of the review, as well as sharing the findings with key stakeholders using platforms such as conferences, and

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presentations through webinars and the HIV Project ECHO (Extension for Community Health Outcomes) (38).

Contributions: FKM conceived the idea, developed the research questions and study methods and contributed meaningfully to the drafting and editing, he also approved the final manuscript. BvW aided in developing the research questions and study methods, contributed meaningfully to the drafting and editing, and approved the final manuscript.

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References

1. Bekker L-G, Siberry GK, Hirnschall G. Ensuring Children and Adolescents Are Not Left Behind [Internet]. 2018 [cited 2020 May 6]. Available from: https://journals.lww.com/jaids

2. UNICEF. HIV and AIDS in Adolescents - UNICEF Data [Internet]. 2020 [cited 2020 Dec 11]. Available from: https://data.unicef.org/topic/adolescents/hiv-aids/

3. Sherr L, Cluver LD, Toska E, He E. Differing psychological vulnerabilities among behaviourally and perinatally HIV infected adolescents in South Africa–implications for targeted health service provision*. AIDS Care - Psychol Socio-Medical Asp AIDS/HIV [Internet]. 2018 Jun 20 [cited 2021 May 25];30:92–101. Available from: https://www.tandfonline.com/doi/abs/10.1080/09540121.2018.1476664

4. UNICEF. Adolescent HIV prevention - UNICEF DATA [Internet]. 2020 [cited 2020 Sep 10]. Available from: https://data.unicef.org/topic/hivaids/adolescents-young-people/

5. UNAIDS. UNAIDS data 2019 [Internet]. 2019 [cited 2020 May 6]. Available from: https://www.unaids.org/sites/default/files/media_asset/2019-UNAIDS-data_en.pdf

6. WHO. WHO | Definition of key terms. WHO [Internet]. 2013 [cited 2016 Oct 11]; Available from: http://www.who.int/hiv/pub/guidelines/arv2013/intro/keyterms/en/

7. UNAIDS. Ending the AIDS epidemic for adolescents, with adolescents: A practical guide to meaningfully engage adolescents in the AIDS response. Geneva; 2016.

8. UNAIDS. Global HIV & AIDS statistics — 2020 fact sheet | UNAIDS [Internet]. 2020 [cited 2020 Dec 12]. Available from: https://www.unaids.org/en/resources/fact-sheet

9. Ssali L, Kalibala S, Birungi J, Egessa A, Wangisi J, Lyavala J, et al. Retention of adolescents living with HIV in care, treatment, and support programs in Uganda. 2014 [cited 2018 Feb 27]; Available from: http://www.hivcore.org/Pubs/Uganda_AdolHAART_Rprt.pdf

10. Grimsrud AT, Pike C, Bekker LG. The power of peers and community in the continuum

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

of HIV care [Internet]. Vol. 8, The Lancet Global Health. Elsevier Ltd; 2020 [cited 2020 Nov 16]. p. e167–8. Available from: https://pubmed.ncbi.nlm.nih.gov/31924538/

11. Kim S-H, Gerver SM, Fidler S, Ward H. Adherence to antiretroviral therapy in adolescents living with HIV: systematic review and meta-analysis. AIDS [Internet]. 2014 Aug 24 [cited 2018 Feb 27];28(13):1945–56. Available from: http://www.ncbi.nlm.nih.gov/pubmed/24845154

12. Nachega JB, Mills EJ, Schechter M. Antiretroviral therapy adherence and retention in care in middle-income and low-income countries: current status of knowledge and research priorities. Curr Opin HIV AIDS [Internet]. 2010 Jan [cited 2018 Sep 30];5(1):70–7. Available from: https://insights.ovid.com/crossref?an=01222929-201001000-00011

13. Ridgeway K, Dulli LS, Murray KR, Silverstein H, Santo LD, Olsen P, et al. Interventions to improve antiretroviral therapy adherence among adolescents in low- and middle-income countries: A systematic review of the literature [Internet]. Vol. 13, PLoS ONE. 2018. 1–33 p. Available from: http://dx.doi.org/10.1371/journal.pone.0189770

14. MacPherson P, Munthali C, Ferguson J, Armstrong A, Kranzer K, Ferrand RA, et al. Service delivery interventions to improve adolescents' linkage, retention and adherence to antiretroviral therapy and HIV care. Trop Med Int Health [Internet]. 2015 Aug [cited 2016 Sep 30];20(8):1015–32. Available from: http://www.ncbi.nlm.nih.gov/pubmed/25877007

15. WHO. WHO | Viral suppression after 12 months of antiretroviral therapy in low- and middle-income countries: a systematic review. WHO [Internet]. 2013 [cited 2017 Jul 13]; Available from: http://www.who.int/bulletin/volumes/91/5/12-112946/en/

16. WHO. WHO | Viral suppression for HIV treatment success and prevention of sexual transmission of HIV [Internet]. 2018 [cited 2020 Dec 12]. Available from: https://www.who.int/hiv/mediacentre/news/viral-supression-hiv-transmission/en/

17. Ridgeway K, Dulli LS, Murray KR, Silverstein H, Santo LD, Olsen P, et al. Interventions to improve antiretroviral therapy adherence among adolescents in low- and middle-income countries: A systematic review of the literature [Internet]. Vol. 13, PLoS ONE. Public Library of Science; 2018 [cited 2020 Jul 9]. Available from:

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

/pmc/articles/PMC5749726/?report=abstract

18. Ammon N, Mason S, Corkery JM. Factors impacting antiretroviral therapy adherence among human immunodeficiency virus-positive adolescents in Sub-Saharan Africa: a systematic review. Public Health [Internet]. 2018 Apr 1 [cited 2018 Sep 30];157:20–31. Available from: http://www.ncbi.nlm.nih.gov/pubmed/29501984

19. Hudelson C, Cluver L. Factors associated with adherence to antiretroviral therapy among adolescents living with HIV/AIDS in low- and middle-income countries: a systematic review. AIDS Care [Internet]. 2015 [cited 2016 Sep 30];27(7):805–16. Available from: http://www.ncbi.nlm.nih.gov/pubmed/25702789

20. Koss CA, Hosek SG, Bacchetti P, Anderson PL, Liu AY, Horng H, et al. Comparison of Measures of Adherence to Human Immunodeficiency Virus Preexposure Prophylaxis Among Adolescent and Young Men Who Have Sex With Men in the United States. Clin Infect Dis An Off Publ Infect Dis Soc Am [Internet]. 2018 Jan 15 [cited 2021 Jul 25];66(2):213. Available from: /pmc/articles/PMC5850042/

21. Casale M, Carlqvist A, Cluver L. Recent Interventions to Improve Retention in HIV Care and Adherence to Antiretroviral Treatment Among Adolescents and Youth: A Systematic Review. AIDS Patient Care STDS [Internet]. 2019 Jun 1 [cited 2020 Apr 26];33(6):237–52. Available from: https://www.liebertpub.com/doi/10.1089/apc.2018.0320

22. Reif LK, Abrams EJ, Arpadi S, Elul B, McNairy ML, Fitzgerald DW, et al. Interventions to Improve Antiretroviral Therapy Adherence Among Adolescents and Youth in Low- and Middle-Income Countries: A Systematic Review 2015–2019 [Internet]. Vol. 1, AIDS and Behavior. Springer; 2020 [cited 2020 Jul 31]. p. 1. Available from: /pmc/articles/PMC7223708/?report=abstract

23. Mavhu W, Willis N, Mufuka J, Bernays S, Tshuma M, Mangenah C, et al. Effect of a differentiated service delivery model on virological failure in adolescents with HIV in Zimbabwe (Zvandiri): a cluster-randomised controlled trial. Lancet Glob Heal [Internet]. 2020 Feb 1 [cited 2020 Apr 26];8(2):e264–75. Available from: https://www.sciencedirect.com/science/article/pii/S2214109X19305261

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

24. Diress G, Dagne S, Alemnew B, Adane S, Addisu A. Viral Load Suppression after Enhanced Adherence Counseling and Its Predictors among High Viral Load HIV Seropositive People in North Wollo Zone Public Hospitals, Northeast Ethiopia, 2019: Retrospective Cohort Study. AIDS Res Treat. 2020;2020.

25. Nasuuna E, Kigozi J, Babirye L, Muganzi A, Sewankambo NK, Nakanjako D. Low HIV viral suppression rates following the intensive adherence counseling (IAC) program for children and adolescents with viral failure in public health facilities in Uganda. BMC Public Health [Internet]. 2018 Dec 22 [cited 2021 Mar 27];18(1):1048. Available from: https://bmcpublichealth.biomedcentral.com/articles/10.1186/s12889-018-5964-x

26. Egger M, Smith GD. Principles of and Procedures for Systematic Reviews. In: Systematic Reviews in Health Care: Meta-Analysis in Context: Second Edition. Wiley Blackwell; 2008. p. 23–42.

27. Müller AD, Jaspan HB, Myer L, Lewis Hunter A, Harling G, Bekker L-G, et al. Standard Measures are Inadequate to Monitor Pediatric Adherence in a Resource-Limited Setting. AIDS Behav [Internet]. 2011 Feb 16 [cited 2016 Oct 10];15(2):422–31. Available from: http://link.springer.com/10.1007/s10461-010-9825-6

28. Grant MJ, Booth A. A typology of reviews: An analysis of 14 review types and associated methodologies [Internet]. Vol. 26, Health Information and Libraries Journal. John Wiley & Sons, Ltd; 2009 [cited 2020 Nov 3]. p. 91–108. Available from: https://onlinelibrary.wiley.com/doi/full/10.1111/j.1471-1842.2009.00848.x

29. Moher D, Shamseer L, Clarke M, Ghersi D, Liberati A, Petticrew M, et al. Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015 statement. Rev Esp Nutr Humana y Diet [Internet]. 2016 [cited 2020 Nov 3];20(2):148–60. Available from: https://pubmed.ncbi.nlm.nih.gov/25554246/

30. Price A, Price, Amy. Mendeley and More for Systematic Reviews. Int Netw Knowl about Wellbeing [Internet]. 2016 Apr 26 [cited 2020 Nov 3]; Available from: http://www.ithinkwell.org/mendeley-and-more-for-systematic-reviews/

31. Mearns H, Otiku PK, Shelton M, Kredo T, Kagina BM, Schmidt BM. Screening strategies

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

for adults with type 2 diabetes mellitus: A systematic review protocol [Internet]. Vol. 9, Systematic Reviews. BioMed Central; 2020 [cited 2020 Nov 16]. p. 156. Available from: https://systematicreviewsjournal.biomedcentral.com/articles/10.1186/s13643-020-01417-3

32. Brignardello-Petersen R, Izcovich A, Rochwerg B, Florez ID, Hazlewood G, Alhazanni W, et al. GRADE approach to drawing conclusions from a network meta-analysis using a partially contextualised framework. BMJ [Internet]. 2020 Nov 10 [cited 2020 Nov 16];371:m3907. Available from: https://www.bmj.com/lookup/doi/10.1136/bmj.m3907

33. Haghighat R, Steinert J, Cluver L. The effects of decentralising antiretroviral therapy care delivery on health outcomes for adolescents and young adults in low- and middle-income countries: a systematic review. Glob Health Action [Internet]. 2019 Jan 1 [cited 2021 Mar 15];12(1):1668596. Available from: https://www.tandfonline.com/doi/full/10.1080/16549716.2019.1668596

34. Mengesha MM, Ajema D, Teshome A, Tura AK. The association between diagnosis disclosure and adherence to antiretroviral therapy among adolescents living with HIV in sub-Saharan Africa: A protocol for systematic review and meta-analysis [Internet]. Vol. 9, Systematic Reviews. BioMed Central; 2020 [cited 2020 Nov 7]. p. 160. Available from: https://systematicreviewsjournal.biomedcentral.com/articles/10.1186/s13643-020-01420-8

35. Egger M, Smith GD, Schneider M, Minder C. Bias in meta-analysis detected by a simple, graphical test. Br Med J [Internet]. 1997 Sep 13 [cited 2020 Nov 16];315(7109):629–34. Available from: https://www.bmj.com/content/315/7109/629

36. Rao G, Lopez-Jimenez F, Boyd J, D'Amico F, Durant NH, Hlatky MA, et al. Methodological standards for meta-analyses and qualitative systematic reviews of cardiac prevention and treatment studies a scientific statement from the American Heart Association. Circulation [Internet]. 2017 Sep 1 [cited 2020 Nov 16];136(10):e172–94. Available from: https://pubmed.ncbi.nlm.nih.gov/28784624/

37. Chaimani A, Mavridis D, Salanti G. A hands-on practical tutorial on performing meta-analysis with Stata. Evid Based Ment Health [Internet]. 2014 Nov 1 [cited 2020 Nov 16];17(4):111–6. Available from: https://pubmed.ncbi.nlm.nih.gov/25288685/

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

38. Bikinesi L, O'Bryan G, Roscoe C, Mekonen T, Shoopala N, Mengistu AT, et al. Implementation and evaluation of a Project ECHO telementoring program for the Namibian HIV workforce. Hum Resour Health [Internet]. 2020 Sep 1 [cited 2020 Nov 17];18(1):1–10. Available from: https://link.springer.com/articles/10.1186/s12960-020-00503-w

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PubMed Results – 23.02.2021 = 64 Results

(adolescents or teenagers or young adults or teen or youth or student or adolescence) AND (antiretroviral therapy) AND (retention) AND (adherence or compliance) AND (viral load) AND (suppression) AND (hiv or aids or acquired human immunodeficiency syndrome or human immunodeficiency virus)

Filters applied: Humans, English, Adolescent: 13-18 years, Adult: 19+ years, Child: 6-12 years, from 2010 - 2020.

Sabinet Results – 23.02.2021 = 68 Results

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[[All: adolescents] OR [All: teenagers] OR [All: young]] AND [[All: adults]
OR [All: teen] OR [All: youth] OR [All: student]] AND [All: or adolescence]
AND [All: antiretroviral] AND [[[All: therapy] AND [All: retention] AND
[All: adherence]] OR [[All: compliance] AND [All: viral]]] AND [All: load]
AND [All: suppression] AND [Languages: English] AND [Publication Date:
(01/01/2010 TO 12/31/2020)]
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ScienceDirect – 08.03.2021 = 834 Results

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adolescents AND antiretroviral therapy AND retention AND adherence OR
compliance AND viral load AND suppression AND hiv or aids AND 10-19 years,
Filters applied: 2010 - 2020
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Scopus – 09.03.2021 = 59 Results

TITLE-ABS-KEY ((adolescents OR teenagers OR young AND adults OR teen OR youth OR student OR adolescence) AND (antiretroviral AND therapy) AND (retention) AND (adherence OR compliance) AND (viral AND load) AND (suppression) AND (hiv OR aids OR acquired AND human AND immunodeficiency AND syndrome OR human AND immunodeficiency AND virus)) Additional Filter: 2010 – 2020

Web of Science – 09.03.2021 = 37 Results

((adolescents or teenagers or young adults or teen or youth or student or adolescence) AND (antiretroviral therapy) AND (retention) AND (adherence or compliance) AND (viral load) AND (suppression) AND (hiv or aids or acquired human immunodeficiency syndrome or human immunodeficiency virus))
Timespan: 2010-2020. **Indexes:** SCI-EXPANDED, SSCI, A&HCI, CPCI-S, CPCI-SSH, BKCI-S, BKCI-SSH, ESCI.

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EBSCOhost – 15.03.2021 = 573 Results

"adolescents or teenagers or young adults or teen or youth AND antiretroviral therapy or antiretroviral treatment AND adherence AND retention AND viral load AND HIV Published Date: 20100101-20201231 AND Apply equivalent subjects on 2021-03-15 10:19 AM"

adolescents or teenagers or young adults or teen or youth AND antiretroviral therapy or antiretroviral treatment AND adherence AND retention AND viral load AND HIV
https://search-ebscohost-com.ezproxy.uwc.ac.za/login.aspx?direct=true&db=rzh&bquery=adolescents+or+teenagers+or+young+adults+or+teen+or+youth+AND+antiretroviral+therapy+or+antiretroviral+treatment+AND+adherence+AND+retention+AND+viral+load+AND+HIV&cli0=DT1&clv0=201001-202012&type=1&searchMode=Relevancy&site=ehost-live&ssl=y

Limiters - Published Date: 20100101-20201231
Expanders - Apply equivalent subjects
Narrow by Language: - english
Narrow by SubjectAge: - child: 6-12 years
Narrow by SubjectAge: - adolescent: 13-18 years
Search modes - SmartText Searchin

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